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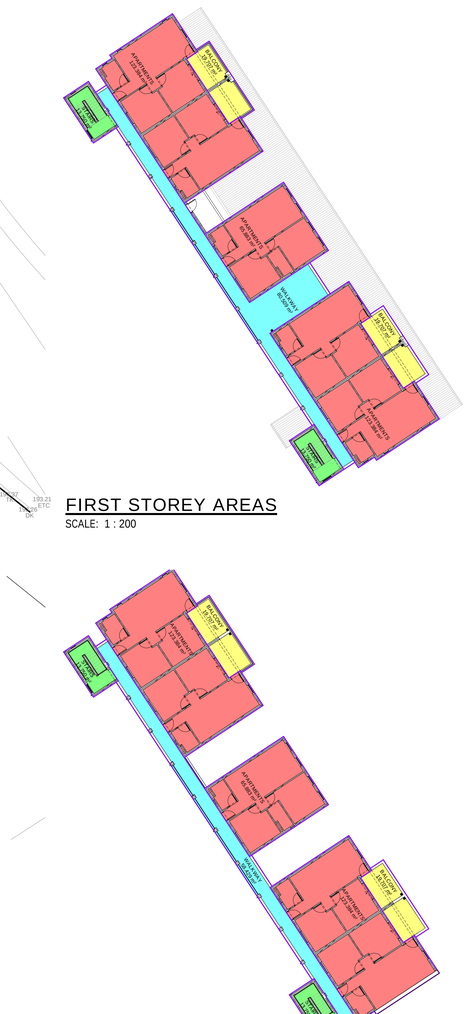
ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

SITE PLAN
SCALE: 1 : 100

GROUND STOREY AREAS
SCALE: 1 : 200



FIRST STOREY AREAS
SCALE: 1 : 200

SECOND STOREY AREAS
SCALE: 1 : 200

Level	GLA Parking Calculations	
	Excluded frm GLA calc.	Included in GLA calc.
Ground Floor	Walkway 183,622m²	Shops 298,324m²
	Toilets 49,936m²	
	Refuse Room 9,094m²	
	Garages 170,345m²	
	Staircases 22,345m²	
	Total Excluded 435,342m²	Total Included 298,324m²
	Parking Requirements	
	Nett:Shop GLA Area	298,324m² @3/100m²
	Parking Required	9bays
	Unit Parking	10units @1,5
	Unit Parking Required	15bays
	Total Parking Required	**24bays**
	Total Parking Provided	**25bays**

Zoning Parameters

Proposed Zoning	: GB1
Permitted Floor Factor	: 1,5
Permitted Coverage	: 100%
Maximum Height	: 15,0m
Street Centreline	: 8,0m
Common Building Lines	: 0,0m

Areas:

Site Area	: 1 735,961m²
Ground Storey Areas	
Shops	: 364,783m²
Stairs	: 27,500m²
Refuse room	: 11,341m²
Garages	: 135,863m²
Covered Walkway	: 174,954m²
TOTAL GROUND STOREY	: 714,441m²
First Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 80,509m²
Covered Balconies	: 39,414m²
TOTAL FIRST STOREY	: 460,074m²
2nd Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 58,429m²
Covered Balconies	: 39,414m²
TOTAL SECOND STOREY	: 437,994m²
Total Construction area	: 1 612,509m²
Permitted Coverage	: 100%
Actual Coverage	: 45%
Permitted Floor Factor	: 1,5
Actual Floor factor	: 0,70
No of Apartments	: 10
No of Shops	: 11
Total Parking Provided:	: 25bays

REFUSE ROOM NOTES

FLOOR
-The floor shall be concrete, screened to a smooth surface & rounded to a minimum height of 75mm around the perimeter.
-The floor shall be finished with a curing agent or washable high gloss floor paint, or alternatively, tiled with durable enamel tiles.

WALLS & ROOF
-Provide adequate protection from the elements and to prevent ingress of rainwater, the whole waste/ recycling storage area/ room shall be walled and roofed.
-Walls shall be constructed of brick, concrete or similar.
-The external finishing must blend in with the material, color scheme & architecture of the main buildings.
-Be rodent proofed in accordance with the Government Rodent proofing regulations.

VENTILATION & LIGHTING
-The waste/ recycling storage area/ room has artificial lighting required.
-The waste/ recycling storage area/ room shall be fitted with solid +/windwood, lockable external doors of which the lower 150mm should have protected sheet metal on the outside. Each door and louvers shall be fitted with an efficient self closing device.
-The door and louvers shall be separated at least 3 meters from any door or window of a habitable room.
-The room has adequate natural ventilation provided by means of fixed louvers.
-Louver doors are fitted and the openings between the louvers are larger than 13mm, then the wire mesh with a diameter size less than 13mm must be fitted over the louver on the inside of the doors.

WATER SUPPLY & DRAINAGE
-Minimum standard hose connection with a tap od no less than 12mm diameter shall be provided in the waste/ recycling storage area/ room for the regular washing of the floor, walls and containers.
-The floor has a drain towards a 100mm trap linked to a drainage pipe discharging to a sewer gully outside the building.
-A grease trap/ gully is provided in terms of the Waste Services Directorate.

ACCESS
-The removal of waste is only affected from the kerbside of a public street and not from entering a private property.
-The recommended location of a communal waster/ recycling areas/ rooms on private property is immediate adjacent to the nearest public road reserve.
-Should the waste/ recycling storage area/ room be locatede at a level different the level of the street entrance to the property, access ramps are to provided to a maximum permissible gradient of 1:7.
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BAYS ON PUBLIC STREET
-Permission for a waste/ recycling storage area/ room will not be allowed next to higher order roads (roads carrying larger volume of traffic) or where traffic flows or traffic sight lines are affected unless a suitable loading bay dedicated by servicing the waste/ recycling storage area/ room is provided by the cost of the developer.
-The minimum requirement is a fully constructed bay with minimum dimensions of 10,5m x 2,5m plus (+) 45 degree splay on either side to allow entrance from and exit on the public street.

NOTES:

CONTAMINATION PROOF TANK
The tank and associated pipework is of contamination proof design and in accordance with the requirements of the Director.

The overflow and vent of a contamination proof tank is screened to prevent the ingress of insects, animals and other sources of pollution.

The contamination proof tank is totally enclosed with no other access to its interior, other than an access panel in its side to facilitate inspection and cleaning, which must be at a level where the tank cannot be used unless the access panel cover is in place; and
the tank and associated pipework is cleaned and maintained in terms of SANS 10252-1.

SIGNAGE
An owner of premises on which non-potable water is used must ensure that every terminal water fitting and every appliance which supplies or uses the water is clearly marked with a weatherproof notice indicating that such water is unsuitable for domestic purposes.

Every warning notice referred to in subsections (1) and (2) must be in the three official languages used in the Province

Awareness for anybody using water at that terminal point (tap) that the water is non-drinkable.

GENERAL
No person may connect a water supply obtained from any source other than the water supply system of the City to any water distribution system without the prior written approval of the Director, and in accordance with any conditions determined by him or her.

COLD WATER DEMAND				
Garden tap	0,0	15L/min		0,0L/min
Whb (mixer)	0	6L/min		00L/min
Sink (mixer)	0	8L/min		00L/min
Wash trough (mixer)	0	6L/min		00L/min
Bath (mixer)	0	25L/min		00L/min
Showerhead	0	7L/min		00L/min
Water Closet (Float valve)	0	5L/min		00L/min
MAXIMUM DEMAND		Q =	00L + 80L =	000L/min

PROBABLE COLD WATER DEMAND				
Garden tap	0	15L/min		00L/min
Whb (mixer)	0	6L/min		00L/min
Sink (mixer)	0	6L/min		00L/min
Wash trough (mixer)	0	6L/min		00L/min
Bath (mixer)	0	25L/min		00L/min
Showerhead	0	7L/min		00L/min
Water Closet (Float valve)	0	5L/min		00L/min
MAXIMUM DEMAND			Q =	00L/min

PROBABLE HOT WATER DEMAND			
in terms of table 5 of SANS 10252-1			
Medium to high rental	NO4	115 to 140L/person / per day	000L/day
Storage volume	NO4	40 to 50L/person / per day	500L
Heater power		2 to 5 kw/ UNIT	

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings.
Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermeable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R. of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2 installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tilted roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	**73**
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II.
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/compartment/a separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the ff to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum Swatts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

SITE PLAN

LEVEL 7
ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax (021) 982 6875
Email. info@level7.co.za
PJ Sylvert STO723

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:36:20	AS SHOWN (A0)	A.H	P.J.S		MU1	211123

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in terms of table 5 of SANS 10252-1

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Awareness for anybody using water at that terminal point (tap) that the water is non-drinkable.

GENERAL
No person may connect a water supply obtained from any source other than the water supply system of the City to any water distribution system without the prior written approval of the Director, and in accordance with any conditions determined by him or her.

GARAGES STOREY FLOOR PLAN

SCALE: 1 : 100

GROUND FLOOR PLAN

SCALE: 1 : 100

SHOP 1

ENERGY FENESTRATION CALCULATION

Internal Area	:	30.358m²
Fenestration Area	:	7.310m²
Percentage	:	24%

More than 20% therefore do not comply
Please refer to Fenestration Rational.

SHOP 2

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 3

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 4

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 5

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 6

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 7

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 8

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 9

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 10

ENERGY FENESTRATION CALCULATION

Internal Area	:	28.085m²
Fenestration Area	:	5.550m²
Percentage	:	20%

20% therefore does comply

SHOP 1

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	30.358m²	100%
Permitted 5% Natural Ventilation Area	:	1.518m²	5%
Actual Natural Ventilation Area	:	3.410m²	11%
Permitted 10% Natural Light Area	:	3.036m²	10%
Actual Natural Light Area	:	7.310m²	24%

SHOP 2

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 3

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 4

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 5

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 6

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 7

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 8

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 9

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 10

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	28.085m²	100%
Permitted 5% Natural Ventilation Area	:	1.404m²	5%
Actual Natural Ventilation Area	:	2.400m²	9%
Permitted 10% Natural Light Area	:	2.809m²	10%
Actual Natural Light Area	:	5.550m²	20%

SHOP 11

NATURAL VENTILATION CALCULATIONS

Actual Room Area	:	30.358m²	100%
Permitted 5% Natural Ventilation Area	:	1.518m²	5%
Actual Natural Ventilation Area	:	3.360m²	11%
Permitted 10% Natural Light Area	:	3.036m²	10%
Actual Natural Light Area	:	7.910m²	26%

ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS:(Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS:(RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermeable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas. Ventilation areas to be min. 5% of habitable room areas. Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy separating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R. of the SANS 1040

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s/m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2 Installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	73
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the flr to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum Swatts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3.7

Roofsheeting	0,03
Rafterspheid:Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183 COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100
Therefore Comply	

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

GROUND STOREY FLOOR PLAN
GARAGE FLOOR PLAN

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:36:26	AS SHOWN (A0)	A.H	P.J.S		MU2	211123

LEVEL 7 ARCHITECTS
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FIRST STOREY FLOOR PLAN

SCALE: 1 : 100

Level	GLA Parking Calculations	
	Excluded frm GLA calc.	Included in GLA calc.
Ground Floor	Walkway 183,622m²	Shops 298,324m²
	Toilets 49,936m²	
	Refuse Room 9,094m²	
	Garages 170,345m²	
	Staircases 22,345m²	
	Total Excluded 435,342m²	Total Included 298,324m²
	Parking Requirements	
	Nett:Shop GLA Area	298,324m² @3/100m²
	Parking Required	9bays
	Unit Parking	10units @1,5
	Unit Parking Required	15bays
	Total Parking Required	**24bays**
	Total Parking Provided	**25bays**

Zoning Parameters

Proposed Zoning	: GB1
Permited Floor Factor	: 1,5
Permited Coverage	: 100%
Maximum Height	: 15,0m
Street Centreline	: 8,0m
Common Building Lines	: 0,0m

Areas:

Site Area	:1 735,961m²
Ground Storey Areas	
Shops	: 364,783m²
Stairs	: 27,500m²
Refuse room	: 11,341m²
Garages	: 135,863m²
Covered Walkway	: 174,954m²
TOTAL GROUND STOREY	: 714,441m²
First Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 80,509m²
Covered Balconies	: 39,414m²
TOTAL FIRST STOREY	: 460,074m²
2nd Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 58,429m²
Covered Balconies	: 39,414m²
TOTAL SECOND STOREY	: 437,994m²
Total Consruction area	: 1 612,509m²
Permitted Coverage	: 100%
Actual Coverage	: 45%
Permited Floor Factor	: 1,5
Actual Floor factor	: 0,70
No of Apartments	: 10
No of Shops	: 11
Total Parking Provided:	: 25bays

ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
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MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

COLD WATER DEMAND

Garden tap	0,0	15L/min		0,0L/min
Wash hand basin	0	6L/min		00L/min
Sink (mixer)	0	6L/min		00L/min
Wash trough (mixer)	0	6L/min		00L/min
Bath (mixer)	0	25L/min		00L/min
Showerhead	0	7L/min		00L/min
Water Closet (Float valve)	0	5L/min		00L/min
MAXIMUM DEMAND		Q =	00L + 80L =	0000L/min

PROBABLE COLD WATER DEMAND

Garden tap	0	15L/min	00L/min
Wha (mixer)	0	6L/min	00L/min
Sink (mixer)	0	6L/min	00L/min
Wash trough (mixer)	0	6L/min	00L/min
Bath (mixer)	0	25L/min	00L/min
Showerhead	0	7L/min	00L/min
Water Closet (Float valve)	0	5L/min	00L/min
MAXIMUM DEMAND			Q = 00L/min

PROBABLE HOT WATER DEMAND

In terms of table 5 of SANS 10252-1			
Medium to high rental	NO4	115 to 140L/person / per day	000L/day
Storage volume	NO4	40 to 50L/person / per day	560L
Heater power		2 to 5 kw/ UNIT	

FIRST STOREY FLOOR PLAN

FENESTRATION CALCULATIONS										
	UNIT 1		UNIT 2		UNIT 3		UNIT 4		UNIT 5	
	int floor area	54,708m²	int floor area	58,439m²	int floor area	56,300m²	int floor area	58,439m²	int floor area	54,708m²
	Fenestration area	10,278m²	Fenestration area	8,768m²	Fenestration area	5,520m²	Fenestration area	8,768m²	Fenestration area	10,878m²
	% fenestration	19%	% fenestration	15%	% fenestration	10%	% fenestration	15%	% fenestration	19%
	Less than 20% therefore does comply		20% therefore does comply		20% therefore does comply		20% therefore does comply		20% therefore does comply	
COLD WATER DEMAND										
Garden Tap	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min
Wash hand basin (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Wash trough (mixer)	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min
Sink (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Bath (mixer)	nod @ 25litre/min	= 0L/min	nod @ 25litre/min	= 0L/min	nod @ 25litre/min	= 0L/min	nod @ 25litre/min	= 0L/min	nod @ 25litre/min	= 0L/min
Showerhead	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min
Water closet (float valve)	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min
Maximum Demand	Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min	
PROBABLE COLD WATER DEMAND										
Garden Tap	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min	nod @ 15 litre/min	= 0L/min
Wash hand basin (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Wash trough (mixer)	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min	nod @ 6 litre/min	= 0L/min
Sink mixer	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Bath (mixer)	nod @ 25 litre/min	= 0L/min	nod @ 25 litre/min	= 0L/min	nod @ 25 litre/min	= 0L/min	nod @ 25 litre/min	= 0L/min	nod @ 25 litre/min	= 0L/min
Showerhead	nod @ 7 litre/min	= 0L/min	nod @ 7 litre/min	= 0L/min	nod @ 7 litre/min	= 0L/min	nod @ 7 litre/min	= 0L/min	nod @ 7 litre/min	= 0L/min
Water closet (float valve)	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min
Probable Demand	Q 2 = 17 litre		Q 2 = 17 litre		Q 2 = 17 litre		Q 2 = 17 litre		Q 2 = 17 litre	
FLATS PROBABLE HOT WATER DEMAND	in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1	
Low rental	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L
Storage volume	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L
Heater power	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW
WATER HEATING										
No of people	4		4		4		4		4	
Usage p/person	65 litres		65 litres		65 litres		65 litres		65 litres	
Daily consumption	260 litres		260 litres		260 litres		260 litres		260 litres	
Ambinent input temp	15 degr		15 degr		15 degr		15 degr		15 degr	
Target output temp	55 degr		55 degr		55 degr		55 degr		55 degr	
Ave temp difference	40 degr		40 degr		40 degr		40 degr		40 degr	
Specific heat for 1L water	4184 kJ/kg		4184 kJ/kg		4184 kJ/kg		4184 kJ/kg		4184 kJ/kg	
Daily energy use	1252kJ		1252kJ		1252kJ		1252kJ		1252kJ	
kWh per day	8,16kWh		8,16kWh		8,16kWh		8,16kWh		8,16kWh	
Annual energy usage	1909 KJ/h		1909 KJ/h		1909 KJ/h		1909 KJ/h		1909 KJ/h	
Efficient target (50%)	4,08 kWh		4,08 kWh		4,08 kWh		4,08 kWh		4,08 kWh	
Solar geyser energy and size	4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)	

FIRST STOREY VENTILATION & LIGHTING AREA CALCULATION

APARTMENT NAME	ROOM NAME	ROOM AREA	PERMITTED (10%) LIGHT AREA	PROVIDED LIGHT AREA	PERMITTED (5%) VENTILATION AREA	PROVIDED VENTILATION AREA
UNIT 1	Lounge & Kitchen	27,428m²	2,743m²	4,513m²	1,374m²	3,413m²
UNIT 1	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 1	Bedroom 2	11,517m²	0,144m²	1,440m²	0,576m²	0,720m²
UNIT 1	Toilet	3,782m²	N/A	0,500m²	N/A	0,250m²
UNIT 2	Lounge & Kitchen	25,246m²	2,525m²	3,263m²	1,262m²	2,413m²
UNIT 2	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 2	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 2	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 3	Lounge & Kitchen	25,328m²	2,534m²	1,800m²	1,266m²	1,800m²
UNIT 3	Master Bedroom	12,715m²	1,273m²	1,800m²	0,636m²	1,800m²
UNIT 3	Master Bedroom Ensuite	3,045m²	N/A	0,240m²	N/A	0,240m²
UNIT 3	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 3	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 4	Lounge & Kitchen	25,246m²	2,525m²	3,263m²	1,262m²	2,413m²
UNIT 4	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 4	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 4	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 5	Lounge & Kitchen	27,428m²	2,743m²	5,113m²	1,374m²	3,363m²
UNIT 5	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 5	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 5	Toilet	3,782m²	N/A	0,500m²	N/A	0,250m²

UNIT 1

ENERGY FENESTRATION CALCULATION

Internal Area	:	54.708m²
Fenestration Area	:	10.278m²
Percentage	:	19%
Less than 20% therefore does comply		

UNIT 2

ENERGY FENESTRATION CALCULATION

Internal Area	:	58.439m²
Fenestration Area	:	8.768m²
Percentage	:	15%
Less than 20% therefore does comply		

UNIT 3

ENERGY FENESTRATION CALCULATION

Internal Area	:	56.300m²
Fenestration Area	:	5.520m²
Percentage	:	10%
Less than 20% therefore does comply		

UNIT 4

ENERGY FENESTRATION CALCULATION

Internal Area	:	58.439m²
Fenestration Area	:	8.768m²
Percentage	:	15%
Less than 20% therefore does comply		

UNIT 5

ENERGY FENESTRATION CALCULATION

Internal Area	:	54.708m²
Fenestration Area	:	10.878m²
Percentage	:	20%
20% therefore does comply		

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermeable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R. of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

POPULATION

SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	**73**
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II.
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. Installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HW/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

Tenancy Separating Elements to have a Min. Fire Resistant of 60m in & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the flr to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard/ Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183 COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100
Therefore Comply	

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

FIRST STOREY FLOOR PLAN

LEVEL 7 ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email. info@level7.co.za
PJ Sylvest ST0723

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:36:33	AS SHOWN (A0)	A.H	P.J.S		MU3	211123

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SECOND STOREY FLOOR PLAN

SCALE: 1 : 100

Level	GLA Parking Calculations	
	Excluded frm GLA calc.	Included in GLA calc.
Ground Floor	Walkway 183,622m²	Shops 298,324m²
	Toilets 49,936m²	
	Refuse Room 9,094m²	
	Garages 170,345m²	
	Staircases 22,345m²	
	Total Excluded 435,342m²	Total Included 298,324m²
	Parking Requirements	
	Nett:Shop GLA Area	298,324m² @3/100m²
	Parking Required	9bays
	Unit Parking	10units @1,5
	Unit Parking Required	15bays
	Total Parking Required	**24bays**
	Total Parking Provided	**25bays**

Zoning Parameters

Proposed Zoning	: GB1
Permited Floor Factor	: 1,5
Permited Coverage	: 100%
Maximum Height	: 15,0m
Street Centreline	: 8,0m
Common Building Lines	: 0,0m

Areas:

Site Area	:1 735,961m²
Ground Storey Areas	
Shops	: 364,783m²
Stairs	: 27,500m²
Refuse room	: 11,341m²
Garages	: 135,863m²
Covered Walkway	: 174,954m²
TOTAL GROUND STOREY	: 714,441m²
First Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 80,509m²
Covered Balconies	: 39,414m²
TOTAL FIRST STOREY	: 460,074m²
2nd Storey Areas	
Apartments	: 312,651m²
Stairs	: 27,500m²
Walkway	: 58,429m²
Covered Balconies	: 39,414m²
TOTAL SECOND STOREY	: 437,994m²
Total Consruction area	: 1 612,509m²
Permitted Coverage	: 100%
Actual Coverage	: 45%
Permited Floor Factor	: 1,5
Actual Floor factor	: 0,70
No of Apartments	: 10
No of Shops	: 11
Total Parking Provided:	: 25bays

ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

COLD WATER DEMAND

Garden tap	0,0	15L/min		0,0L/min
Sink (mixer)	0	6L/min		00L/min
Sink (mixer)	0	6L/min		00L/min
Wash trough (mixer)	0	6L/min		00L/min
Bath (mixer)	0	25L/min		00L/min
Showerhead	0	7L/min		00L/min
Water Closet (Float valve)	0	5L/min		00L/min
MAXIMUM DEMAND			Q = 00L + 80L =	000L/min

PROBABLE COLD WATER DEMAND

Garden tap	0	15L/min		00L/min
Sink (mixer)	0	6L/min		00L/min
Sink (mixer)	0	6L/min		00L/min
Wash trough (mixer)	0	6L/min		00L/min
Bath (mixer)	0	25L/min		00L/min
Showerhead	0	7L/min		00L/min
Water Closet (Float valve)	0	5L/min		00L/min
MAXIMUM DEMAND			Q =	00L/min

PROBABLE HOT WATER DEMAND

in terms of table 5 of SANS 10252-1			
Medium to high rental	NO4	115 to 140L/person / per day	000L/day
Storage volume	NO4	40 to 50L/person / per day	560L
Heater power		2 to 5 kw/ UNIT	

SECOND STOREY FLOOR PLAN

FENESTRATION CALCULATIONS												
	UNIT 6		UNIT 7		UNIT 8		UNIT 9		UNIT 10	
	int floor area	54,708m²	int floor area	58,439m²	int floor area	56,300m²	int floor area	58,439m²	int floor area	54,708m²
	Fenestration area	10,278m²	Fenestration area	8,768m²	Fenestration area	5,520m²	Fenestration area	8,768m²	Fenestration area	10,878m²
	% fenestration	19%	% fenestration	15%	% fenestration	10%	% fenestration	15%	% fenestration	19%
	Less than 20% therefore does comply		20% therefore does comply		20% therefore does comply		20% therefore does comply		20% therefore does comply	
COLD WATER DEMAND										
Garden Tap	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min
Wash hand basin (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Wash trough (mixer)	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min
Sink (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Bath (mixer)	no0 @ 25litre/min	= 0L/min	no0 @ 25litre/min	= 0L/min	no0 @ 25litre/min	= 0L/min	no0 @ 25litre/min	= 0L/min	no0 @ 25litre/min	= 0L/min
Showerhead	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min	no1 @ 7 litre/min	= 7L/min
Water closet (float valve)	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min
Maximum Demand	Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min		Q = 24 litre + 80 litre = 104 litre/min	
PROBABLE COLD WATER DEMAND										
Garden Tap	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min	no0 @ 15 litre/min	= 0L/min
Wash hand basin (mixer)	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Wash trough (mixer)	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min	no0 @ 6 litre/min	= 0L/min
Sink mixer	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min	no1 @ 6 litre/min	= 6L/min
Bath (mixer)	no0 @ 25 litre/min	= 0L/min	no0 @ 25 litre/min	= 0L/min	no0 @ 25 litre/min	= 0L/min	no0 @ 25 litre/min	= 0L/min	no0 @ 25 litre/min	= 0L/min
Showerhead	no0 @ 7 litre/min	= 0L/min	no0 @ 7 litre/min	= 0L/min	no0 @ 7 litre/min	= 0L/min	no0 @ 7 litre/min	= 0L/min	no0 @ 7 litre/min	= 0L/min
Water closet (float valve)	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min	no1 @ 5 litre/min	= 5L/min
Probable Demand	Q 2 + 17 litre		Q 2 + 17 litre		Q 2 + 17 litre		Q 2 + 17 litre		Q 2 + 17 litre	
FLATS PROBABLE HOT WATER DEMAND	in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1		in terms of table 5 of SANS 10252-1	
Low rental	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L	65 to 75L/ person	= 260L
Storage volume	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L	20 to 25L/person	= 80L
Heater power	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW	2 to 3 kW/unit	= 3 kW
WATER HEATING										
No of people	4		4		4		4		4	
Usage p/person	65 litres		65 litres		65 litres		65 litres		65 litres	
Daily consumption	260 litres		260 litres		260 litres		260 litres		260 litres	
Ambient input temp	15 degr		15 degr		15 degr		15 degr		15 degr	
Target output temp	55 degr		55 degr		55 degr		55 degr		55 degr	
Ave temp difference	40 degr		40 degr		40 degr		40 degr		40 degr	
Specific heat for 1L water	4184 kJ/kg		4184 kJ/kg		4184 kJ/kg		4184 kJ/kg		4184 kJ/kg	
Daily energy use	1252kJ		1252kJ		1252kJ		1252kJ		1252kJ	
kWh per day	8,16kWh		8,16kWh		8,16kWh		8,16kWh		8,16kWh	
Annual energy usage	1909 kJh		1909 kJh		1909 kJh		1909 kJh		1909 kJh	
Efficient target (50%)	4,08 kWh		4,08 kWh		4,08 kWh		4,08 kWh		4,08 kWh	
Solar geyser energy and size	4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)		4,5 kWh (200L Solar comply)	

SECOND STOREY VENTILATION & LIGHTING AREA CALCULATION

APARTMENT NAME	*ROOM NAME*	*ROOM AREA*	*PERMITTED (10%) LIGHT AREA*	*PROVIDED LIGHT AREA*	*PERMITTED (5%) VENTILATION AREA*	*PROVIDED VENTILATION AREA*
UNIT 6	Lounge & Kitchen	27,428m²	2,743m²	4,513m²	1,374m²	3,413m²
UNIT 6	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 6	Bedroom 2	11,517m²	0,144m²	1,440m²	0,576m²	0,720m²
UNIT 6	Toilet	3,782m²	N/A	0,500m²	N/A	0,250m²
UNIT 7	Lounge & Kitchen	25,246m²	2,525m²	3,263m²	1,262m²	2,413m²
UNIT 7	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 7	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 7	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 8	Lounge & Kitchen	25,328m²	2,534m²	1,800m²	1,266m²	1,800m²
UNIT 8	Master Bedroom	12,715m²	1,273m²	1,800m²	0,636m²	1,800m²
UNIT 8	Master Bedroom Ensuite	3,045m²	N/A	0,240m²	N/A	0,240m²
UNIT 8	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 8	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 9	Lounge & Kitchen	25,246m²	2,525m²	3,263m²	1,262m²	2,413m²
UNIT 9	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 9	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 9	Toilet	3,695m²	N/A	0,240m²	N/A	0,240m²
UNIT 10	Lounge & Kitchen	27,428m²	2,743m²	5,113m²	1,374m²	3,363m²
UNIT 10	Master Bedroom	11,981m²	1,198m²	3,825m²	0,600m²	1,913m²
UNIT 10	Bedroom 2	11,517m²	1,152m²	1,440m²	0,576m²	0,720m²
UNIT 5	Toilet	3,782m²	N/A	0,500m²	N/A	0,250m²

UNIT 6

ENERGY FENESTRATION CALCULATION

Internal Area	:	54.708m²
Fenestration Area	:	10.278m²
Percentage	:	19%
Less than 20% therefore does comply		

UNIT 7

ENERGY FENESTRATION CALCULATION

Internal Area	:	58.439m²
Fenestration Area	:	8.768m²
Percentage	:	15%
Less than 20% therefore does comply		

UNIT 8

ENERGY FENESTRATION CALCULATION

Internal Area	:	56.300m²
Fenestration Area	:	5.520m²
Percentage	:	10%
Less than 20% therefore does comply		

UNIT 9

ENERGY FENESTRATION CALCULATION

Internal Area	:	58.439m²
Fenestration Area	:	8.768m²
Percentage	:	15%
Less than 20% therefore does comply		

UNIT 10

ENERGY FENESTRATION CALCULATION

Internal Area	:	54.708m²
Fenestration Area	:	10.878m²
Percentage	:	20%
20% therefore does comply		

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermiable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R. of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. Installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only Should airconditioning or under floor heating to be installed.

POPULATION

SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	**73**
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the flr to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

SECOND STOREY FLOOR PLAN

LEVEL 7 ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email: info@level7.co.za
PJ Sylver STOT23

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:36:37	AS SHOWN (A0)	A.H	P.J.S		MU4	211123

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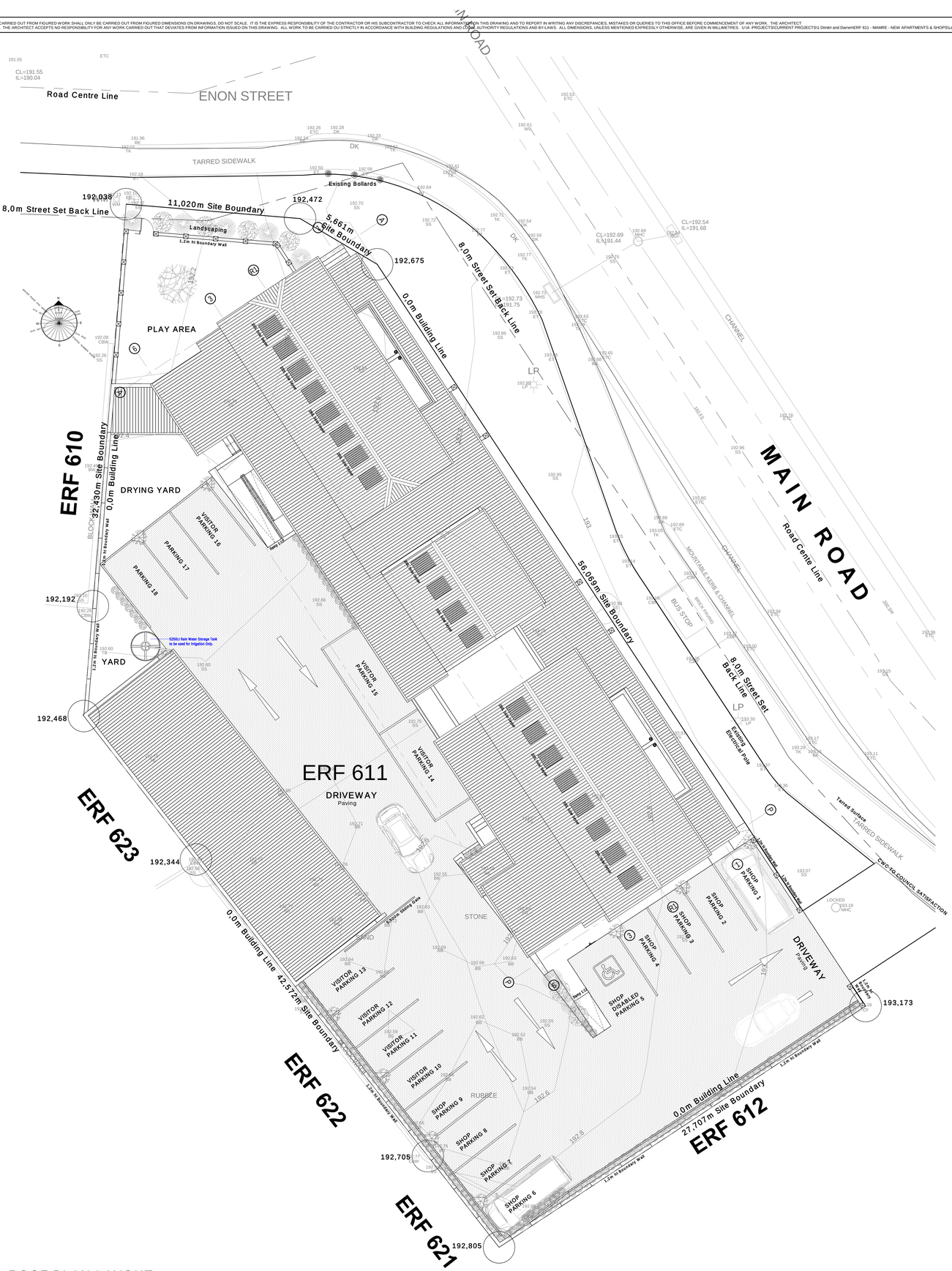
MAIN ROAD
Road Centre Line
ENON STREET
Road Centre Line
CL=191.55
IL=190.04
TARRED SIDEWALK
DK
Existing Bollards
192,038
192,472
192,675
192,192
192,468
192,344
192,705
192,805
193,173
11,020m Site Boundary
8,0m Street Set Back Line
5,661m Site Boundary
0,0m Building Line
56,069m Site Boundary
27,707m Site Boundary
0,0m Building Line 42,572m Site Boundary
ERF 610
27,430m Site Boundary
0,0m Building Line
ERF 611
DRIVEWAY
Paving
ERF 623
ERF 622
ERF 621
ERF 612
Landscaping
1,2m ht Boundary Wall
PLAY AREA
DRYING YARD
YARD
VISITOR PARKING 18
PARKING 17
PARKING 18
VISITOR PARKING 15
VISITOR PARKING 14
VISITOR PARKING 13
VISITOR PARKING 12
VISITOR PARKING 11
VISITOR PARKING 10
SHOP PARKING 9
SHOP PARKING 8
SHOP PARKING 7
SHOP PARKING 6
SHOP PARKING 5
SHOP PARKING 4
SHOP PARKING 3
SHOP PARKING 2
SHOP PARKING 1
SHOP DISABLED PARKING 5
Bay Paving
Bay Paving
Bay Paving

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NORTH EAST ELEVATION

SCALE: 1 : 100

NORTH ELEVATION

SCALE: 1 : 100

SOUTH EAST ELEVATION

SCALE: 1 : 100

NORTH WEST ELEVATION

SCALE: 1 : 100

SOUTH WEST ELEVATION

SCALE: 1 : 100

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1.8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermeable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

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I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. Installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOOR/SLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	73
MIN 1000 wide escape width	

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the ffl to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

ELEVATIONS

LEVEL 7 ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 5875
Email: info@level7.co.za
PJ Sytsma STO723

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:36:56	AS SHOWN (A0)	A.H	P.J.S		MU6	211123

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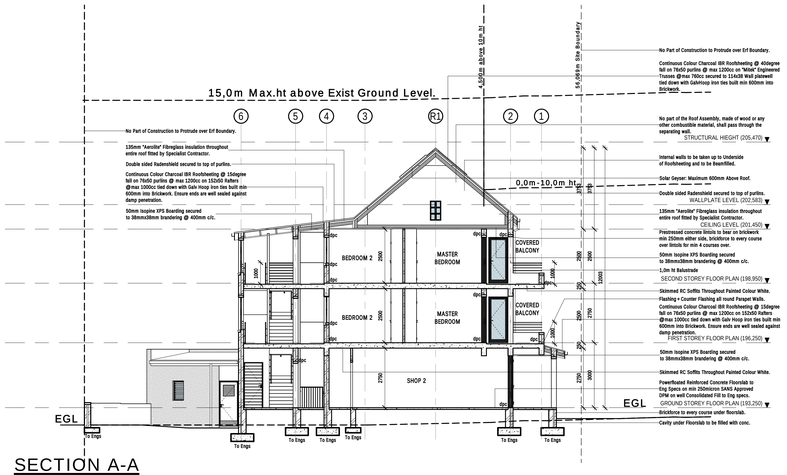

SECTION A-A
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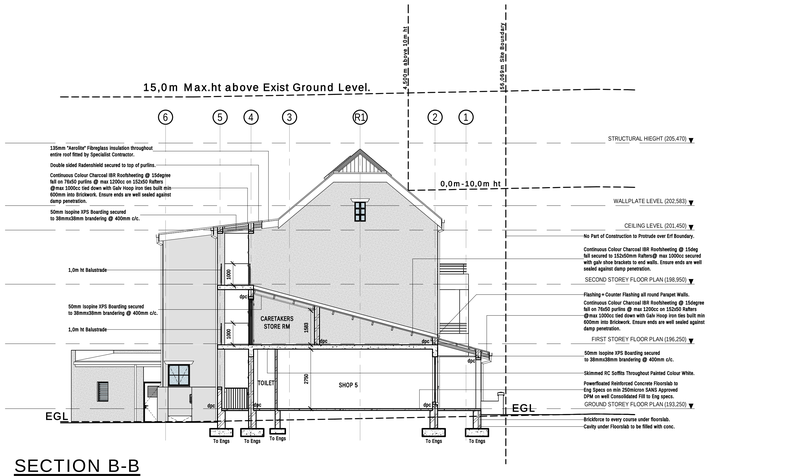

SECTION B-B
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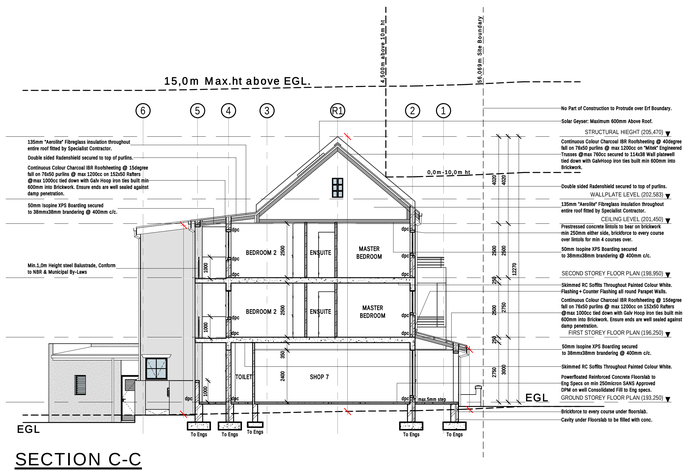
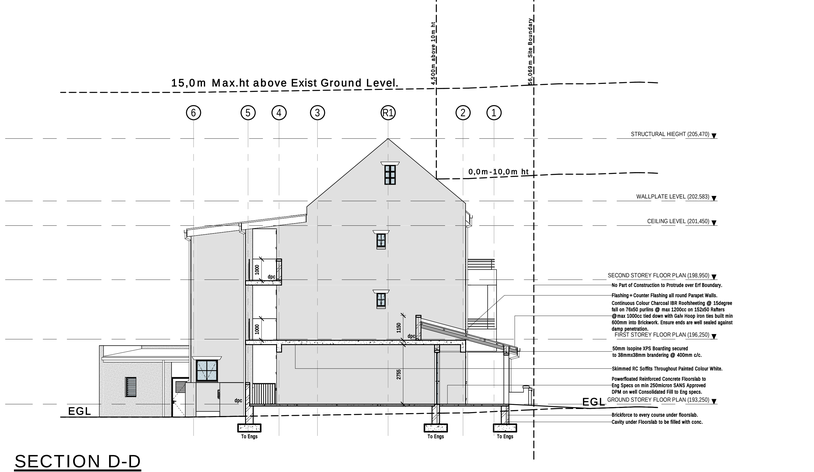


SECTION C-C
SCALE: 1 : 100



SECTION D-D
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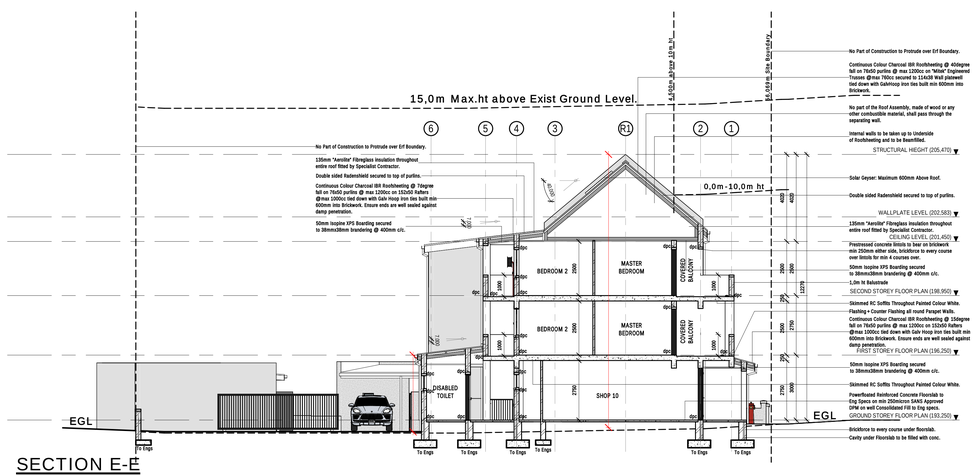

SECTION E-E
SCALE: 1 : 100

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS:(Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS:(RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermiable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have disposal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R. of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II.
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. Installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	**20**
FIRST STOREY FLOOR PLAN	**20**
GROUND STOREY FLOOR PLAN	**33**
Total Population	**73**
MIN 1000 wide escape width	

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the ffl to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radenshield Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

SECTIONS A-E

LEVEL 7
ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email: info@level7.co.za
PJ Sylvest STOTZ3

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:37:02	AS SHOWN (A0)	A.H	P.J.S		MU7	211123

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SECTION F-F
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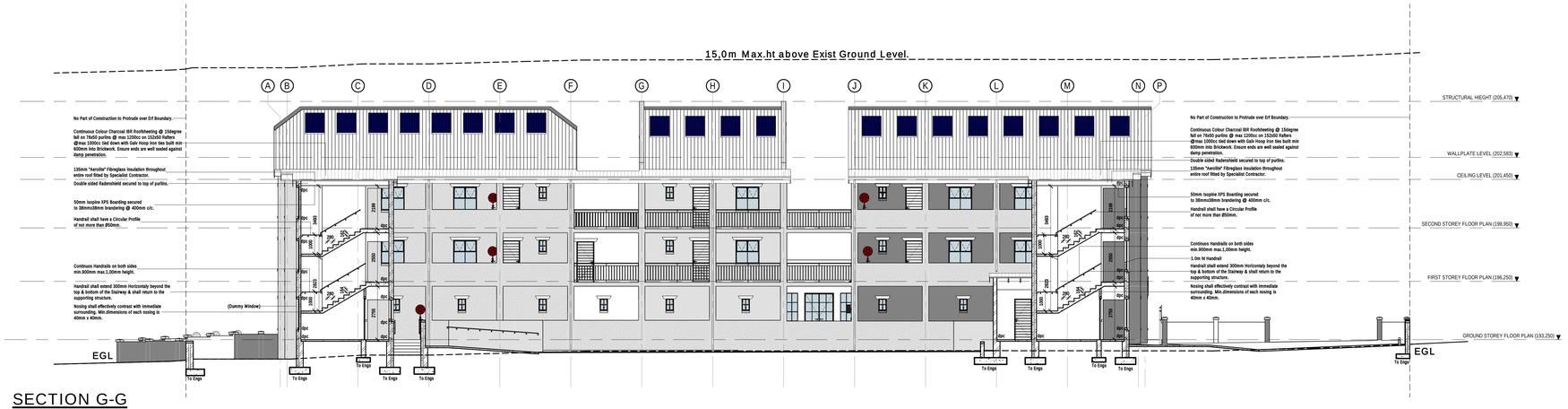

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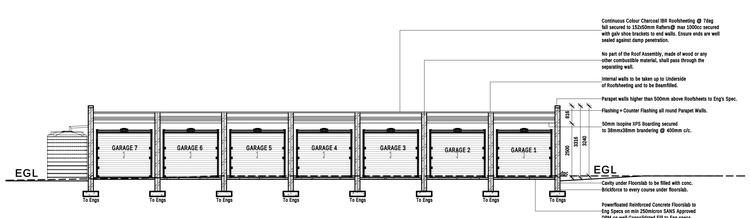

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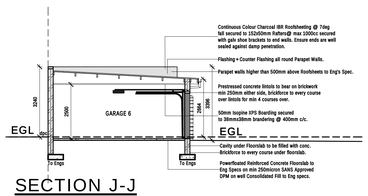

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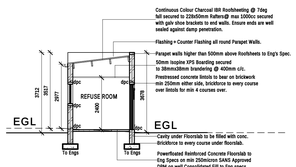

SECTION K-K
SCALE: 1 : 100

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1.8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustraders - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermiable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light area to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissable air leakage (AL) for openable glazed units shall be less than 2,0Ls - m2 and 0,31Ls-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0Ls - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	73
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with Sans 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the ff to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3.7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

SECTIONS F-K

LEVEL 7
ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email. info@level7.co.za
PJ Syfret ST0723

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:37:09	AS SHOWN (A0)	A.H	P.J.S		MU8	211123

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Existing Bollards
11,020m Site Boundary
0,0m Building Line
Landscaping
1,2m ht Boundary Wall
PLAY AREA
32,430m Site Boundary
5,661m Site Boundary
8,0m Street Set Back Line
0,0m Building Line
LANDSCAPING
56,069m Site Boundary
9,0m Street Set Back Line
0,0m Building Line
42,572m Site Boundary
27,707m Site Boundary
0,0m Building Line
ERF 610
DRYING YARD
ERF 611
DRIVEWAY
Paving
ERF 623
ERF 622
ERF 621
ERF 612
SHOP 1
SHOP 2
SHOP 3
SHOP 4
SHOP 5
SHOP 6
SHOP 7
SHOP 8
SHOP 9
SHOP 10
SHOP 11
SHOP 12
SHOP DISABLED PARKING 5
SHOP PARKING 1
SHOP PARKING 2
SHOP PARKING 3
SHOP PARKING 4
SHOP PARKING 6
SHOP PARKING 7
SHOP PARKING 8
SHOP PARKING 9
VISITOR PARKING 10
VISITOR PARKING 11
VISITOR PARKING 12
VISITOR PARKING 13
VISITOR PARKING 14
VISITOR PARKING 15
VISITOR PARKING 16
PARKING 17
PARKING 18
GARAGE 1
GARAGE 2
GARAGE 3
GARAGE 4
GARAGE 5
GARAGE 6
GARAGE 7
SERVICE ROOM
Occupancy F2 3 People
Occupancy F2 30 People
Occupancy F2 2 People
Occupancy F2 1 People
Occupancy F2 0 People
Occupancy F2 10 People
Occupancy F2 5 People
Occupancy F2 6 People
Occupancy F2 4 People
Occupancy F2 7 People
Occupancy F2 8 People
Occupancy F2 9 People
Occupancy F2 11 People
Occupancy F2 12 People
Occupancy F2 13 People
Occupancy F2 14 People
Occupancy F2 15 People
Occupancy F2 16 People
Occupancy F2 17 People
Occupancy F2 18 People
Occupancy F2 19 People
Occupancy F2 20 People
Occupancy F2 21 People
Occupancy F2 22 People
Occupancy F2 23 People
Occupancy F2 24 People
Occupancy F2 25 People
Occupancy F2 26 People
Occupancy F2 27 People
Occupancy F2 28 People
Occupancy F2 29 People
Occupancy F2 31 People
Occupancy F2 32 People
Occupancy F2 33 People
Occupancy F2 34 People
Occupancy F2 35 People
Occupancy F2 36 People
Occupancy F2 37 People
Occupancy F2 38 People
Occupancy F2 39 People
Occupancy F2 40 People
Occupancy F2

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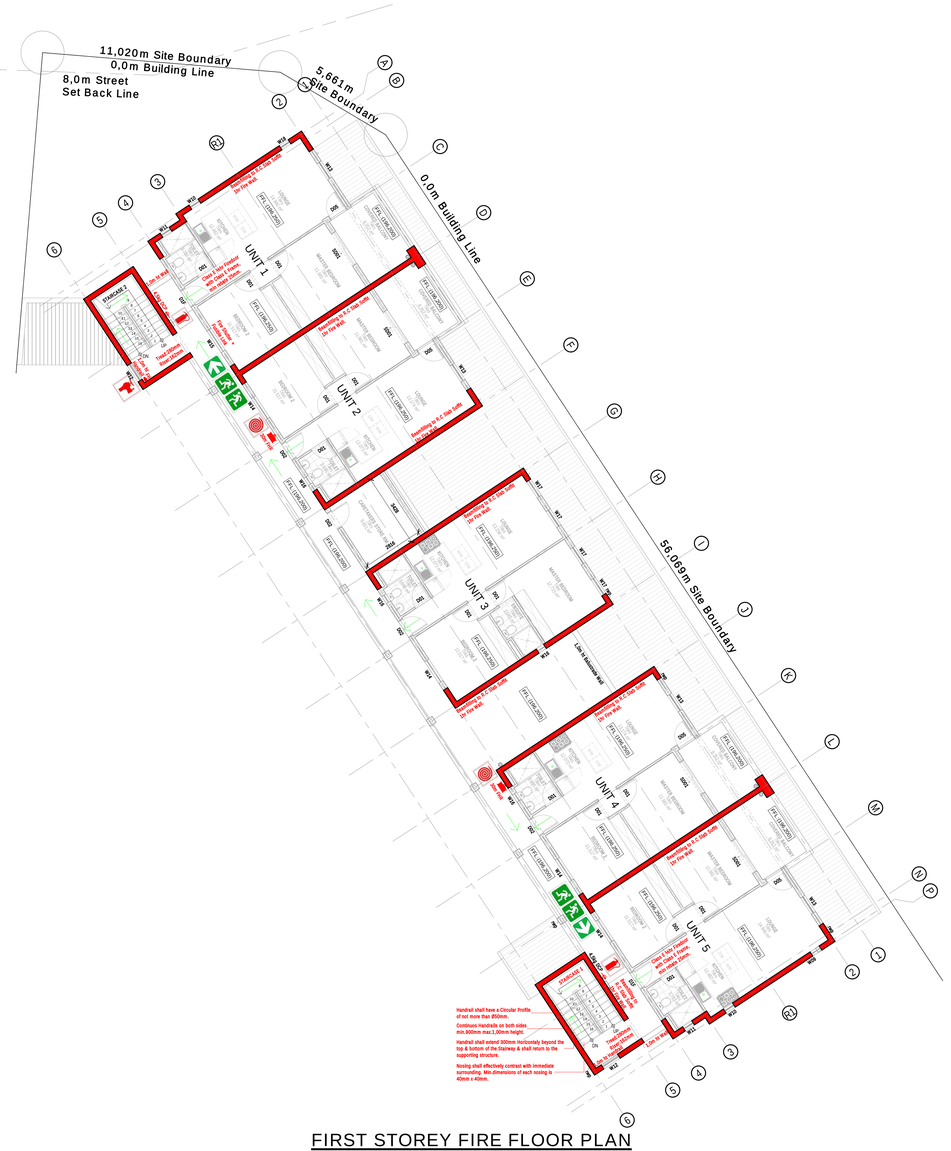
SECOND STOREY FIRE FLOOR PLAN

SCALE: 1 : 100



FIRST STOREY FIRE FLOOR PLAN

SCALE: 1 : 100

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings. Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermeable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light areas to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	73
MIN 1000 wide escape width	

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II.
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

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Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3,7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100

Therefore Comply

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NEW APARTMENTS+SHOPS

FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

FIRST & SECOND STOREY FIRE FLOOR PLAN

LEVEL 7
ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email. info@level7.co.za
PJ Syfret ST0723

DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:37:23	AS SHOWN (A0)	A.H	P.J.S		MU10	211123

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8,0m Street Set Back Line
11,020m Site Boundary
0,0m Building Line
32,430m Site Boundary
5,661m Site Boundary
8,0m Street Set Back Line
0,0m Building Line
56,089m Site Boundary
8,0m Street Set Back Line
0,0m Building Line
27,707m Site Boundary
ERF 610
ERF 611
DRIVEWAY
Paving
ERF 623
ERF 622
ERF 621
ERF 612
DRYING YARD
YARD
SHOP 1
SHOP 2
SHOP 3
SHOP 4
SHOP 5
SHOP 6
SHOP 7
SHOP 8
SHOP 9
SHOP 10
SHOP 11
REFUSE ROOM
GARAGE 7
GARAGE 6
GARAGE 5
GARAGE 4
GARAGE 3
GARAGE 2
GARAGE 1
VISITOR PARKING 16
PARKING 17
PARKING 18
VISITOR PARKING 15
VISITOR PARKING 14
VISITOR PARKING 13
VISITOR PARKING 12
VISITOR PARKING 11
VISITOR PARKING 10
SHOP PARKING 1
SHOP PARKING 2
SHOP PARKING 3
SHOP PARKING 4
SHOP PARKING 5
SHOP PARKING 6
SHOP PARKING 7
SHOP PARKING 8
SHOP PARKING 9
SHOP DISABLED PARKING 5
LANDSCAPING
DRIVEWAY
Paving
12mm 3ϕ Tap
Fit Water Saver Flow Restrictor 12L/min
Fit Water Saver Flow Restrictor 12L/min
Fit Water Saver Flow Restrictor 12L/min

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11,020m Site Boundary
0,0m Building Line
8,0m Street
Set Back Line
5,661m Site Boundary
0,0m Building Line
56,069m Site Boundary
UNIT 1
UNIT 2
UNIT 3
UNIT 4
UNIT 5
UNIT 6
UNIT 7
UNIT 8
UNIT 9
UNIT 10
150mm FLR
200mm AKT
FIRE HYDRANT
Fire Hydrant Sign
Fire Hydrant Sign
Fire Hydrant Sign

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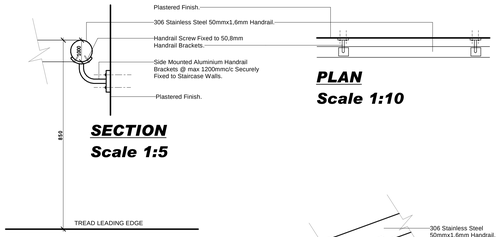
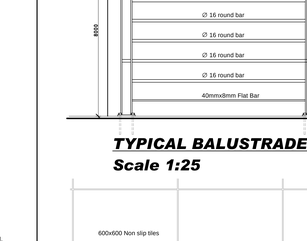
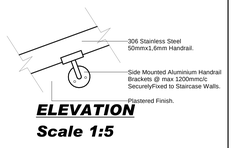


SECTION
Scale 1:5

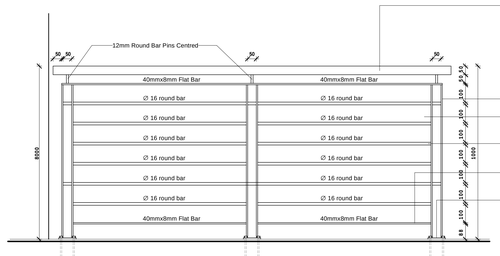
TYPICAL STAIR HANDRAIL



PLAN
Scale 1:10

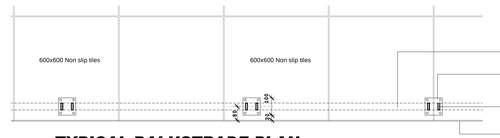


ELEVATION
Scale 1:5

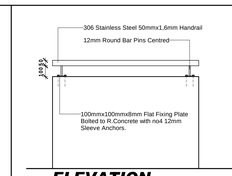


TYPICAL BALUSTRADE ELEVATION
Scale 1:25

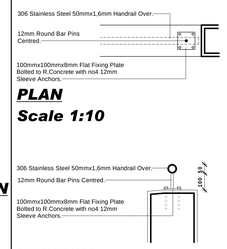
TYPICAL BALUSTRADE SECTION
Scale 1:25



TYPICAL BALUSTRADE PLAN
Scale 1:25



ELEVATION
Scale 1:20

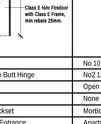
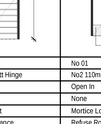
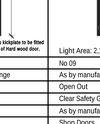
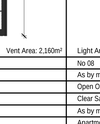
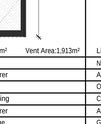
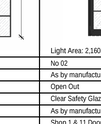
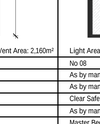
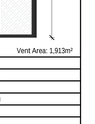
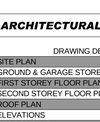


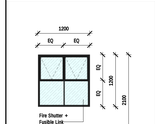
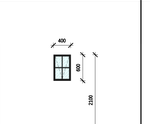
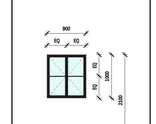
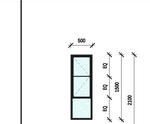
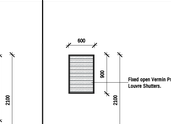
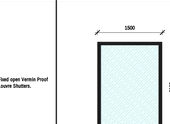
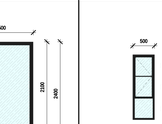
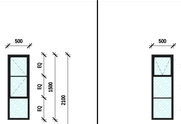
PLAN
Scale 1:10

SECTION
Scale 1:5

EXTERNAL HANDRAIL

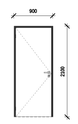
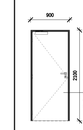
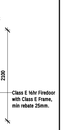
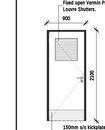
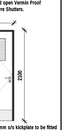
BALUSTRADE DETAILING

	W01	W02	W03	W04 - (Dummy Window)	W05	W06	W07	W08	W09	W10	W11	W12	W13	W14
														
	Light Area: 3,150m² Vent Area: 0,000m²	Light Area: 0,750m² Vent Area: 0,500m²	Light Area: 0,500m² Vent Area: 0,250m²	Light Area: 0,000m² Vent Area: 0,000m²	Light Area: 0,240m² Vent Area: 0,240m²	Light Area: 1,000m² Vent Area: 1,000m²	Light Area: 1,350m² Vent Area: 1,350m²	Light Area: 1,350m² Vent Area: 0,450m²	Light Area: 1,350m² Vent Area: 0,450m²	Light Area: 0,500m² Vent Area: 0,250m²	Light Area: 0,500m² Vent Area: 0,250m²	Light Area: 1,350m² Vent Area: 1,350m²	Light Area: 1,350m² Vent Area: 0,900m²	Light Area: 1,440m² Vent Area: 0,720m²
NO OFF	No 09	No 02	No 02	No 03	No 10	No 01	No 01	No 01	No 02	No 03	No 04	No 04	No 08	No 08
HINGES	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
CILL	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
GLAZING	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing
LOCKSET	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
LOCATION	Shop Fronts	Shop 1	Toilet Shop 1	Staircase 2	Shop Toilets	Staircase	Staircase Ground	Shop 11	Apartment Lounge	Apartment Kitchen	Apartment Toilets	Staircases	Apartment Lounge	Bedroom 2
COLOUR	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum
SHGC VALUE	-	0,330	0,330	-	-	-	-	-	-	-	-	-	-	-
NOTES														

	W15	W16	W17	W18	W19	W20	W21	W22
								
	Light Area: 1,440m² Vent Area: 0,720m²	Light Area: 0,240m² Vent Area: 0,240m²	Light Area: 0,900m² Vent Area: 0,900m²	Light Area: 0,750m² Vent Area: 0,500m²	Light Area: 0,540m² Vent Area: 0,540m²	Light Area: 3,150m² Vent Area: 0,000m²	Light Area: 0,750m² Vent Area: 0,500m²	Light Area: 0,500m² Vent Area: 0,250m²
NO OFF	No 02	No 08	No 08	No 02	No 01	No 02	No 01	No 02
HINGES	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
CILL	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
GLAZING	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing
LOCKSET	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
LOCATION	Bedroom	Toilet	Master Bedroom & Lounge	Lounge	Refuse Room	Shop Fronts	Shop 11	Toilet Shop 11
COLOUR	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum	Colour Charcoal epoxy coated alum
SHGC VALUE	-	-	-	-	-	0,356	-	-
NOTES	Fire Shutter + Fusible Link							

WINDOW SCHEDULE

SCALE: 1 : 50

	D01	D1F	D02	D03	D04	D05	D06	D07	SD01
									
	Light Area: 2,160m² Vent Area: 2,160m²	Light Area: 1,913m² Vent Area:1,913m²	Light Area: 6,547m² Vent Area: 3,600m²	Light Area: 2,160m² Vent Area: 2,160m²	Light Area: 3,825m² Vent Area: 1,913m²				
NO OFF	No 41	No 01	No 10	No 01	No 09	No 08	No 01	No 02	No 08
HINGES	No2 110mm Butt Hinge	No3 110mm Butt Hinge	No2 110mm Butt Hinge	No2 110mm Butt Hinge	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
CILL	No Cill	Open In	Open In	Open In	Open Out	Open Out	Open Out	Open Out	As by manufacturer
GLAZING	None	None	None	None	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing	Clear Safety Glazing
LOCKSET	Mortice Lockset	Mortice Lockset	Mortice Lockset	Mortice Lockset	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer
LOCATION	Bedrooms & Toilets	Apartment Entrance	Apartment Entrance	Refuse Room	Shop Doors	Apartment Lounge	Ground Entrace Door	Shop 1 & 11 Doors	Master Bedroom
COLOUR	hdwd painted	As by Manufacturer Painted	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	As by manufacturer	Colour Charcoal epoxy coated alum
SHGC VALUE	-	-	-	-	-	-	-	0,289	-
NOTES	Hollowcore Door	Class E \ohr Firedoor with Class E Frame, min rebate 25mm.	Solidcore Door						

ARCHITECTURAL DRAWING LIST

SHEET	DRAWING DESCRIPTION	PAPER SIZE
MU1	SITE PLAN	(A0)
MU2	GROUND & GARAGE STOREY FLOOR PLAN	(A0)
MU3	FIRST STOREY FLOOR PLAN	(A0)
MU4	SECOND STOREY FLOOR PLAN	(A0)
MU5	ROOF PLAN	(A0)
MU6	ELEVATIONS	(A0)
MU7	SECTIONS A-E	(A0)
MU8	SECTIONS F-K	(A0)
MU9	GROUND STOREY FIRE FLOOR PLAN	(A0)
MU10	FIRST & SECOND STOREY FIRE FLOOR PLAN	(A0)
MU11	GROUND STOREY HOT WATER FLOOR PLAN	(A0)
MU12	FIRST & SECOND STOREY HOT WATER FLOOR PLAN	(A0)
MU13	HANDRAIL DETAIL & SCHEDULES	(A0)

NOTES:
Figured dimensions to be used in preference to scaling. All building work to be strictly in accordance with SANS 0400 & local authority requirements.

FOUNDATIONS:
All footings as per Engineers drawings.

SURFACE BEDS & RC SLABS:
All surface beds and RC slabs as per Engineers drawings.
Surface beds on 250 micron DPM on 50mm sand blinding layer on well compacted fill.

WALLS (Brickwork):
External walls - 230mm cavity walls unless shown otherwise.
Internal walls - 190mm & 90mm solid walls.
External cavity walls to have stepped DPC's at floor level & around all openings.
Cavities to be concrete filled to u/s stepped DPC at ground level.
Weepholes @ 1000mm c.c. to all stepped DPC's.
Prestressed concrete lintols over all openings in brickwork where no RC beams specified with minimum of 4 courses of brickwork over laid to manufacturers specification.
Tie cavity walls together with 5 galvanized butterfly ties per square metre.

WALLS (RC Concrete):
Reinforced concrete walls to Engineers Detail.
Reinforced concrete walls to have vertical waterproofing with subsoil drain.

WALL & FLOOR FINISHES:
Floor finishes - as indicated on plan.
Wall finishes - tiling full height to bathrooms, toilets & kitchens. Plaster & paint elsewhere.

DOORS & WINDOWS:
As per door & windows schedules.

GLAZING:
All glazing to be carried out in accordance with Part N of the SANS National Building Regulations.
All glazing to comply with requirements of SANS 10137 & SANS 10400 part N - access doors and side lights to have safety glass. Windows lower than 500mm from floor, windows lower than 1800mm above pitch line of stairs and shop fronts to be safety glass.
WINDOWS IN EXCESS OF 1sqm TO BE SAFETY GLAZING.
WINDOWS WITHIN 1,8m FROM A BATH OR SHOWER CUBICLE TO BE SAFETY GLAZING.

HANDRAILS:
All handrails to be 1m high.
Handrails to Staircase to be 900mm High.
Staircases - Galvanised handrail on steel structure bolted to face of concrete stair.
Balustrades - Balustrades may not permit the passage of a 100mm diameter ball.

CONCRETE ROOF:
Sea or Riverstone as per specialists on waterpermiable filter fabric by specialist on 50mm thick isoboard.inverted flat roof
insulation on no2 coats ABE uni-torch on waterproofing membranes on min 25mm screed to slope to roof outlets at min 1:100 on RC concrete roofslab to Eng specs.

STRUCTURAL STEEL:
All structural steelwork in roof cavity to Eng. detail.

LIGHT & VENTILATION:
Light areas to be min. 10% of habitable room areas.
Ventilation areas to be min. 5% of habitable room areas.
Mechanical ventilation for internal bathrooms & toilets as per SANS standards.

Fire dampner complying with SANS 1193 shall be fitted wherever it breaches a occupancy seperating element.

PLUMBING & DRAINAGE:
All sewer pipes running under slabs or footings to be encased in 150mm well compacted soil all round.
All waste fittings to have deepseal traps.
All sewer pipes to be 110mm Ø PVC at minimum fall of 1:80.
All waste pipes to enter separately into soil & vent stack.
All plumbing work to be in accordance with council regulations and materials to carry the relevant SANS mark.
Gutters - 80mm u PVC-gutters.
Downpipes - 80mm diam PVC.

STORMWATER:
Stormwater disposal is to comply with Part R of the SANS 10400

GENERAL:
Balustrade material shall comply with SANS 0104
All structural walls between individual units shallcomply with SANS 0177, part 2.

H3 OCCUPANCY

FIRE PROTECTION

A. Fire Extinguishers

Be SANS approved units- serviced once per annum by a registered holder of a certificate of competence. Installed, maintained and serviced in accordance with the relevant SANS codes of practice. The position thereof to be clearly indicated
by symbolic signs to comply with SANS 1186.

B. Hose Reels

So located that the end of the hose can reach any point within the protected area. Shall be fitted with the reel disc facing the user.Discharge water at a minimum of 0,5 l/sec at a minimum pressure of 300kPa. Be 30m SANS 543 approved.

C. Hydrants

Shall be fitted with a 16mm internal diam nozzle, to comply with SANS 1128: PART I and part II
Discharge water at a minimum flow rate of 20 l/sec at a minimum pressure of 300kPa and a maximum 700kPa.

D. Fire Water Supply

The supply of water for a fire installation must be in accordance with the provisionsof SANS 10252-1.

E. Suppression Equipment

Shall be clearly visible and where not, be indicated by means of SANS approved safety symbolic signage.

F. Escape Routes

Shall lead on ground floor directly to a street or approved open space. Be indicated directly above exits with SANS approved signage as being an emergency escape route

G. Signage - Emergency/Escape routes

SANS approved (running man and directional arrows) to be adequately provided to clearly indicate direction of travel for escape purposes. Escape route path to be kept clear and unobstructed at all times.

H. Emergency Lighting

Shall be independent of the normal mains supply and function for a minimum period 60 or 120 minutes, whichever is required, in the event of normal failure. Escape routes shall be provided with artificial lighting at a minimum illuminance of 50lux, 100mm above the floor. Fitted above exit doors or doors leading to stairways. Emergence lights and minimum light levels on escape routes shall comply with SANS 10114-2 and SANS 1462-22.

I. Fire Rated Doors

Shall be fitted with an automatic self-closing device.

J. Fire Stopping

Any services passing through (horizontal or vertical) a fire division/occupancy separating element shall be adequately fire stopped.

K. External sliding gates to be locked with padlocks to ensure that
Fire and Rescue services are able to cut the lock in event of emergency

L. Deviations

No deviations from the notations as indicated on the the plan or stipulated requirements, permitted.

GENERAL:
Escape Routes to be clearly marked and signposted to indicate direction of travel.

Any mark or sign shall comply with SANS 1186.

Fire extinguishers to comply with SANS 810, SANS 889 or SANS 1151,
and shall be installed and serviced in accordance with SANS 0105.

No smoking signs of approved size shall be prominently displayed, and shall be in accordance with the requirements in SANS 1186.

In any building, the controls of any lift shall be so designed that, in the event of fire, such lift shall be brought automatically to the main entrance storey without stopping and shall remain there with its doors open

ENERGY NOTES:

WINDOWS: as part of fenestration
Permissible air leakage (AL) for openable glazed units shall be less than 2,0L/s - m2 and 0,31L/s-m2 for non openable units, when tested in accordance with SANS 613.

SWING DOORS/ REVOLVING DOORS (fenestration)
The max AL shall be 5,0L/s - m2 and a draught protection device shall be fitted to the bottom edge of each door leaf.

EXTERNAL DOORS
a foamer, rubber or compressible strip seal to restrict airflow shall be fitted to each and every edge of an external door

EXHAUST FANS
Shall be fitted with a self closing damper or when fitted in any habitable room.

HOT WATER SUPPLY
To be supplied by 150L Solar Water Heating System, that complies with SANS 1307 and SANS 10106, based on thermal performance in accordance with SANS 6211-1 and SANS 6211-2. installation to comply with SANS 10254 fitted by specialist contractor.
Solar Geyser system shall be maintained in accordance with SANS 10252-1.

Hot water vessels/tanks shall be insulated with a material achieving an R-value not less than 2,0 if a full solar is not used. Solar water heating system shall comply with SANS 1307, SANS 10106, SANS 10254 and SANS 10252-1
Min 80mm fibreglass blanket to be fitted to hotwater reservoirs.

Piping: All exposed pipes to and from the HWC/central heating system shall be insulated with pipe insulation with a minimum R-value of 1 including cold water supply and pressure relief piping within 1m from the heating device. min 30mm XPS insulation around hot water pipes

ROOF INSULATION
100mm fibreglass insulation to be fitted. All joints to be properly sealed.
Wall plate and roof junctions shall be sealed.
Tiled roofs shall have an underlay and joints shall be sealed. Joints in sheeted roofs shall be sealed

LIGHTING
Minimum lighting shall be in accordance with SANS 10114-1 and SANS 10400 - O

ENERGY DEMAND
Maximum energy demand shall be 5W/m2 and the max energy consumption shall be 5kWh/m2

FLOORSLAB INSULATION
Either 30mm thick XPS or 25mm PU foam insulation to be used under concrete floorslabs to habitable areas only should airconditioning or under floor heating to be installed.

POPULATION	
SECOND STOREY FLOOR PLAN	20
FIRST STOREY FLOOR PLAN	20
GROUND STOREY FLOOR PLAN	33
Total Population	73
MIN 1000 wide escape width	

Tenancy Separating Elements to have a Min. Fire Resistant of 60min & that it will extend to the underside of the Slab or Roof Covering.

Glazing less than 500mm from the ffl to be safety glazing.

Balustrades may not permit the passage of a 100mm diameter ball.

All Fire and Safety wall to be raised to opening Height.

Lighting: Maximum 5watts/m²

Demand Management System:
- Schedule of water efficient showerheads & taps to be used
- Water efficient dual flush cistern to be used
- Dual flush cistern with a 3 ltr and 6 ltr flush.
- Taps with maximum flow rate of 6 ltr/min.
- Shower heads with max. flow rate of 7ltr/min.
- Demand at wash hand basin.

Water Conservation:
Storage Tank To Capture Non-Portable Rainwater, to be used for Irrigational Purposes only.

Fire signs shall be of photo luminescent type and shall comply with requirements set in SANS 1186-1, 1186-3, 1186-5 and 1186-22. Signs shall comply with SANS 10114-2 with regard to maximum viewing distance.

Fire Hydrant to comply with SANS 1128-1.

The Carriageway Crossing must be constructed in accordance with the City's specifications for Carriageway Crossings.

ROOF R-VALUES

Climatic zone 4 - UP

REQUIRED R VALUE= 3.7

Roofsheeting	0,03
Radiensheild Double Sided	1,57
50mm XPS Isoboard Ceiling	2,083
Flexible fibreglass (75-100mm) (density 10 to 18 kg/m3)	2,50
TOTAL R VALUE	6,183
	COMPLY

WALL R-VALUES

Brick Cavity Wall - Comply with deemed to satisfaction

CR-VALUES

CR value of wall	112
Required	= 100
Therefore Comply	

HOT WATER

Minimum 50% of Hot Water Supplied by the Solar Geyser.

NEW APARTMENTS+SHOPS

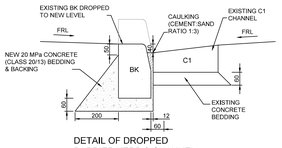
FOR MR C.B SCHOEMAN
ON ERF 611 MAMRE

MUNICIPAL APPROVAL:

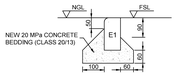
HANDRAIL DETAIL & SCHEDULES

LEVEL 7 ARCHITECTS
11 Graafland Street, Brackenfell 7560
PO Box 712, Parow, 7499
Tel 021 982 2642 - Fax: (021) 982 6875
Email. info@level7.co.za
PJ Syfret ST0723

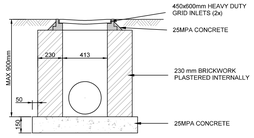
DATE	SCALE:	DRAWN:	CHECKED:	REVISION NO.	DRAWING NO.	PROJECT NO.
12 Nov 2025 11:37:40	AS SHOWN (A0)	A.H	P.J.S		MU13	211123



DETAIL OF DROPPED BARRIER KERB & CHANNEL AT CARRIAGEWAY CROSSING
SCALE 1:10



DETAIL OF FOOTWAY EDGING
SCALE 1:10



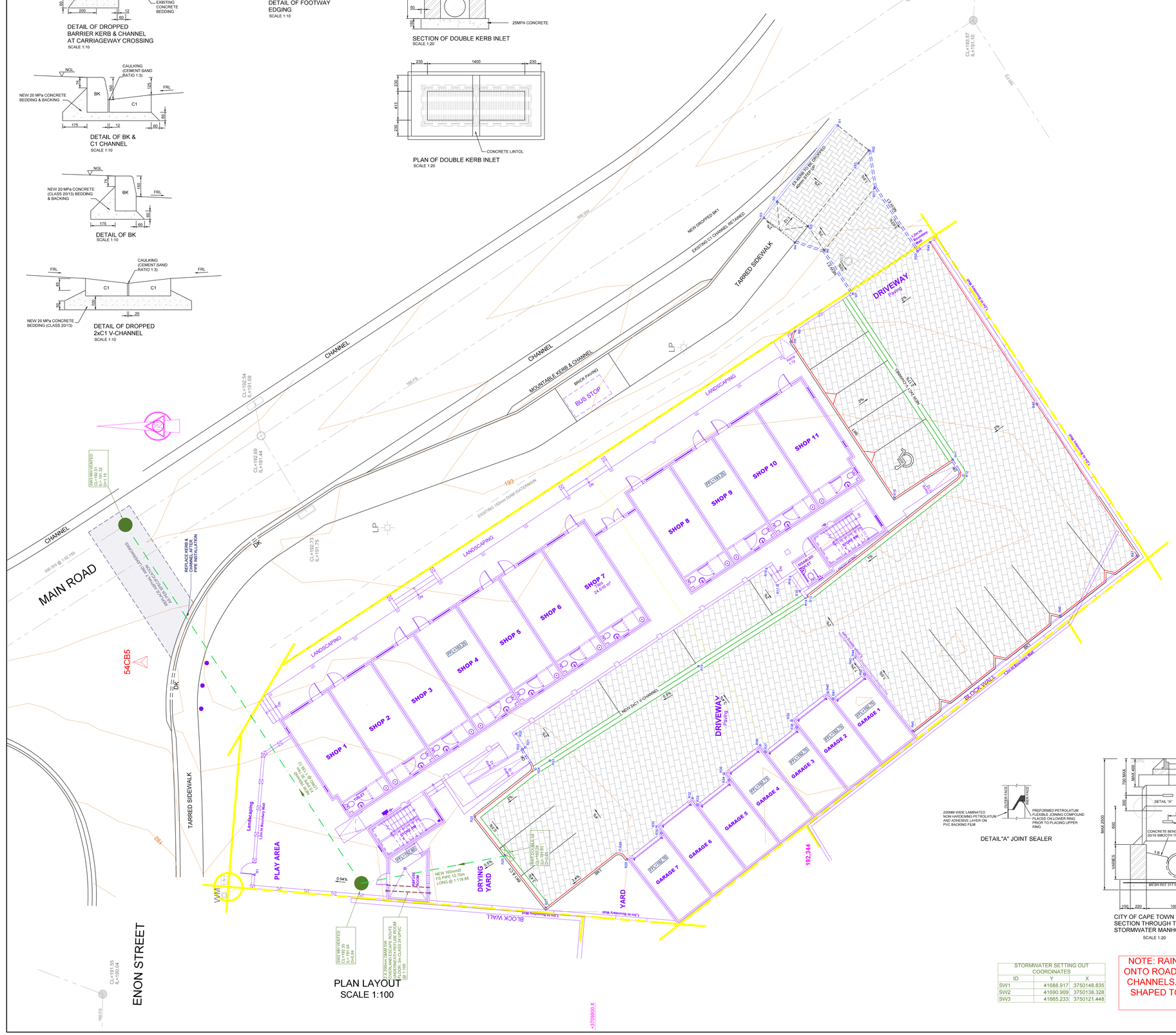
SECTION OF DOUBLE KERB INLET
SCALE 1:20

DETAIL OF BK & C1 CHANNEL
SCALE 1:10

PLAN OF DOUBLE KERB INLET
SCALE 1:20

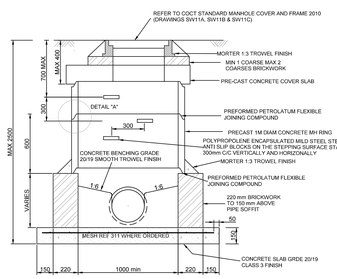
DETAIL OF BK
SCALE 1:10

DETAIL OF DROPPED 2xC1 V-CHANNEL
SCALE 1:10



ROAD SETTING OUT COORDINATE TABLE			
ID	Y	X	FRL
R1	41636.616	3750172.475	193.30
R2	41642.059	3750168.120	193.23
R3	41643.230	3750167.183	193.30
R4	41645.010	3750169.408	193.37
R5	41643.839	3750170.345	193.29
R6	41645.578	3750171.511	193.35
R7	41648.685	3750173.592	193.00
R8	41651.353	3750169.412	193.09
R9	41652.117	3750169.175	193.10
R10	41663.303	3750176.540	192.80
R11	41660.553	3750180.716	192.70
R12	41661.686	3750181.462	192.68
R13	41670.379	3750170.339	192.54
R14	41670.521	3750170.157	192.54
R15	41669.984	3750169.738	192.55
R16	41668.969	3750169.069	192.59
R17	41669.574	3750168.150	192.55
R18	41668.379	3750167.364	192.60
R19	41675.355	3750162.501	192.48
R20	41680.194	3750149.437	192.47
R21	41681.178	3750150.092	192.44
R22	41681.304	3750149.900	192.44
R23	41682.118	3750151.858	192.42
R24	41682.719	3750150.831	192.41
R25	41686.136	3750146.459	192.30
R26	41688.913	3750148.629	192.28
R27	41692.675	3750151.570	192.54
R28	41688.533	3750156.871	192.67
R29	41689.202	3750157.394	192.70
R30	41687.306	3750159.821	192.70
R31	41686.841	3750159.458	192.70
R32	41684.803	3750162.066	192.70
R33	41685.268	3750162.429	192.73
R34	41683.513	3750164.675	192.73
R35	41683.048	3750164.311	192.73
R36	41681.009	3750166.919	192.73
R37	41681.474	3750167.283	192.75
R38	41679.719	3750169.528	192.75
R39	41679.254	3750169.165	192.75
R40	41677.216	3750171.773	192.75
R41	41677.681	3750172.136	192.75
R42	41675.926	3750174.382	192.75
R43	41674.855	3750173.544	192.65
R44	41674.713	3750173.725	192.65
R45	41679.992	3750177.851	192.75
R46	41671.734	3750188.417	192.92
R47	41667.435	3750193.918	193.05
R48	41656.593	3750186.729	192.84
R49	41644.984	3750179.085	193.14
R50	41645.660	3750178.186	193.11
R51	41641.075	3750175.119	193.41
R52	41638.396	3750174.700	193.38
R53	41639.332	3750173.952	193.38

DETAIL"A" JOINT SEALER



CITY OF CAPE TOWN TYPICAL DETAIL SW3
SECTION THROUGH TYPICAL STORMWATER MANHOLE
SCALE 1:20

STORMWATER SETTING OUT COORDINATES		
ID	Y	X
SW1	41688.917	3750148.835
SW2	41690.909	3750138.328
SW3	41665.233	3750121.448

NOTE: RAINWATER PIPES TO DISCHARGE ONTO ROADWAYS. ROADWAYS SHAPED TO CHANNELS. EMERGENCY ESCAPE ROUTE SHAPED TOWARDS NORTHERN CORNER TO ENON STREET.

NOTES

GENERAL:
1. CONSTRUCTION MUST BE DONE IN STRICT ACCORDANCE WITH SABS 1200 AND GENERAL CONDITIONS OF CONTRACT (1990).
2. DRAWINGS ARE NOT TO BE SCALED FROM AND QUERIES SHOULD BE DIRECTED DIRECTLY TO THE ENGINEER OR HIS APPOINTED REPRESENTATIVE.
3. THE POSITION OF ALL EXISTING SERVICES TO BE CONFIRMED ON SITE BEFORE CONSTRUCTION COMMENCES. ALL EXISTING SERVICES TO BE LOCATED ON DRAWINGS OR POINTED OUT ON SITE BY THE ENGINEER OR MUNICIPAL OFFICIALS. THAT ARE DAMAGED ARE TO BE REPAIRED AT THE CONTRACTOR'S EXPENSE TO THE STANDARD ACCEPTED BY THE ENGINEER OR OWNER OF SAID SERVICES (TELKOM, ESKOM ETC) STANDARDS.
4. THE CONTRACTOR MUST OBTAIN ALL RELEVANT WAYLEAVES & PERMITS FROM THE LOCAL AUTHORITY PRIOR TO ANY EXCAVATION TAKING PLACE ON SITE. IT IS ACCEPTED THAT THE CONTRACTOR HAS MADE ALLOWANCE IN HIS RATES FOR THE WAYLEAVES AND ANY DEPOSITS. NO ADDITIONAL PAYMENT WILL BE MADE FOR WAYLEAVES.

STORMWATER:
1. ALL CONCRETE STORMWATER PIPES TO BE PRECAST CONCRETE, CLASS 100D SPIGOT & SOCKET LAID ON "CLASS B" BEDDING AS PER SABS1200.D
2. ALL PVC PIPES TO BE PVC CLASS 34 (HEAVY DUTY UNDERGROUND SEWER) LAID ON "FLEXIBLE TYPE" BEDDING AS PER SABS 1200.B
3. BEDDING AND BLANKET MATERIAL MAY ONLY BE SELECTED FROM TRENCH EXCAVATION IF INSTRUCTED BY THE ENGINEER. SELECTED MATERIAL MUST BE CLEAN AND FREE OF CLAY, STONES AND ROCKS. ENGINEER TO BE NOTIFIED BEFORE INSTALLING ANY PIPES. OTHERWISE ALL BEDDING AND BLANKET MUST BE IMPORTED FROM A COMMERCIAL SOURCE.
4. ALL MANHOLES TO BE AS PER COCT STANDARD DETAIL SW5.
5. ALL GRID INLETS TO BE AS PER STANDARD DETAIL
6. ALL RATES UNDER EXCAVATION IN THE BILLS TO INCLUDE BACKFILLING (EXCLUDING BEDDING AND BLANKET AS PER SABS1200.B). THE REMOVAL OF EXCESS MATERIAL FROM SITE, AND THE COMPACTION OF BACKFILL TO "UNDER ROADS" STANDARD AS DESCRIBED IN SABS1200DB. SHOULD THE COMPACTION OF ANY TRENCH BE IN DOUBT, THE ENGINEER WILL ORDER THAT THE COMPACTION BE TESTED AND PROVEN AT THE COST OF THE CONTRACTOR. NO ADDITIONAL PAYMENT WILL BE MADE FOR SUCH TESTS.
7. THE RATE FOR MANHOLES OR CATCHPITS WILL INCLUDE BACKFILLING AND COMPACTION AROUND MANHOLES. NO ADDITIONAL PAYMENT WILL BE MADE FOR THIS.

ROADS:
1. LAYERWORKS OF CARRIAGEWAY CROSSING & REFUSE EMBAYMENT TO CONSIST OF THE FOLLOWING:
   - 80mm THICK CONCRETE PAVER CLASS 40/2.6 ACCORDING TO SANS 1058 ANNEXURE A ON 25mm THICK BEDDING SAND
   - 150mm C4 CEMENT STABILISED SUBBASE CBR > 80 COMPACTED TO 96% MOD AASHTO
   - 150mm G7 SSG CBR >15 COMPACTED TO 98% MOD AASHTO (100% FOR SAND)
   - 150mm IN-SITU ROADBED PREPARATION COMPACTED TO 93 %MOD AASHTO (100% FOR SAND)
2. LAYERWORKS OF INTERNAL ROADS TO CONSIST OF THE FOLLOWING:
   - 60mm THICK CONCRETE PAVER CLASS 40/2.6 ACCORDING TO SANS 1058 ANNEXURE A ON 25mm THICK BEDDING SAND
   - 150mm G5 SUBBASE > 45 COMPACTED TO 95% MOD AASHTO
   - 150mm G7 SSG CBR >15 COMPACTED TO 98% MOD AASHTO (100% FOR SAND)(ONLY IF ORDERED BY THE ENGINEER)
   - 150mm IN-SITU ROADBED PREPARATION COMPACTED TO 93 %MOD AASHTO (100% FOR SAND)
3. RATES FOR MATERIALS SHALL COVER ALL TESTING. NO ADDITIONAL PAYMENT WILL BE MADE FOR TESTING. TESTING FREQUENCY AS PER SANS 1200.
4. KERBS AND CHANNELS TO BE INSTALLED AS PER DETAILS.

100mm ON ORIGINAL

LEGEND

○ R1	ROAD SETTING OUT POINTS
	NEW PAVING IN CARRIAGEWAY CROSSING (SEE NOTES FOR DETAILS)
	NEW PAVING FOR ROADWAYS ON SITE (SEE NOTES FOR DETAILS)
	RE-INSTATE ASPHALT - LAYERS TO MATCH EXISTING
	NEW STORMWATER MANHOLE
	NEW 2 x 600 X 400mm CI STORMWATER GRID INLET
DK	NEW DROPPED KERB
	EXISTING SEWER PIPE
	EXISTING SEWER MANHOLE
	EXISTING STORMWATER PIPE
	EXISTING STORMWATER MANHOLE
	EXISTING STORMWATER CATCHPIT
	EXISTING WATER PIPE
	EXISTING FIRE HYDRANT
	EXISTING VALVE

FOR APPROVAL

No.	AMENDMENT	DATE

CLIENT
AZARIA PROPERTY DEVELOPERS

PROJECT
NEW DEVELOPMENT ON ERF 611 MAMRE

DRAWING TITLE
CIVIL ENGINEERING SERVICES
ROADS & STORMWATER

SIGNED: CARSTENS CONSULTING ENGINEERS
....., ..., Pr Eng nR 200570006

DATE (ORIGINAL)	DRAWN	CHECKED
1-8-2025	CLR	TJC

DRAWING NO			REV NO	SCALE
CACE540	C	300	0	AS SHOWN

CARSTENS CONSULTING ENGINEERING
E202 De Ville Centre, Wellington Way
Durbanville, 7550, South Africa
M +27 82 410 3558
E tzolw@cace.co.za
www.cace.co.za

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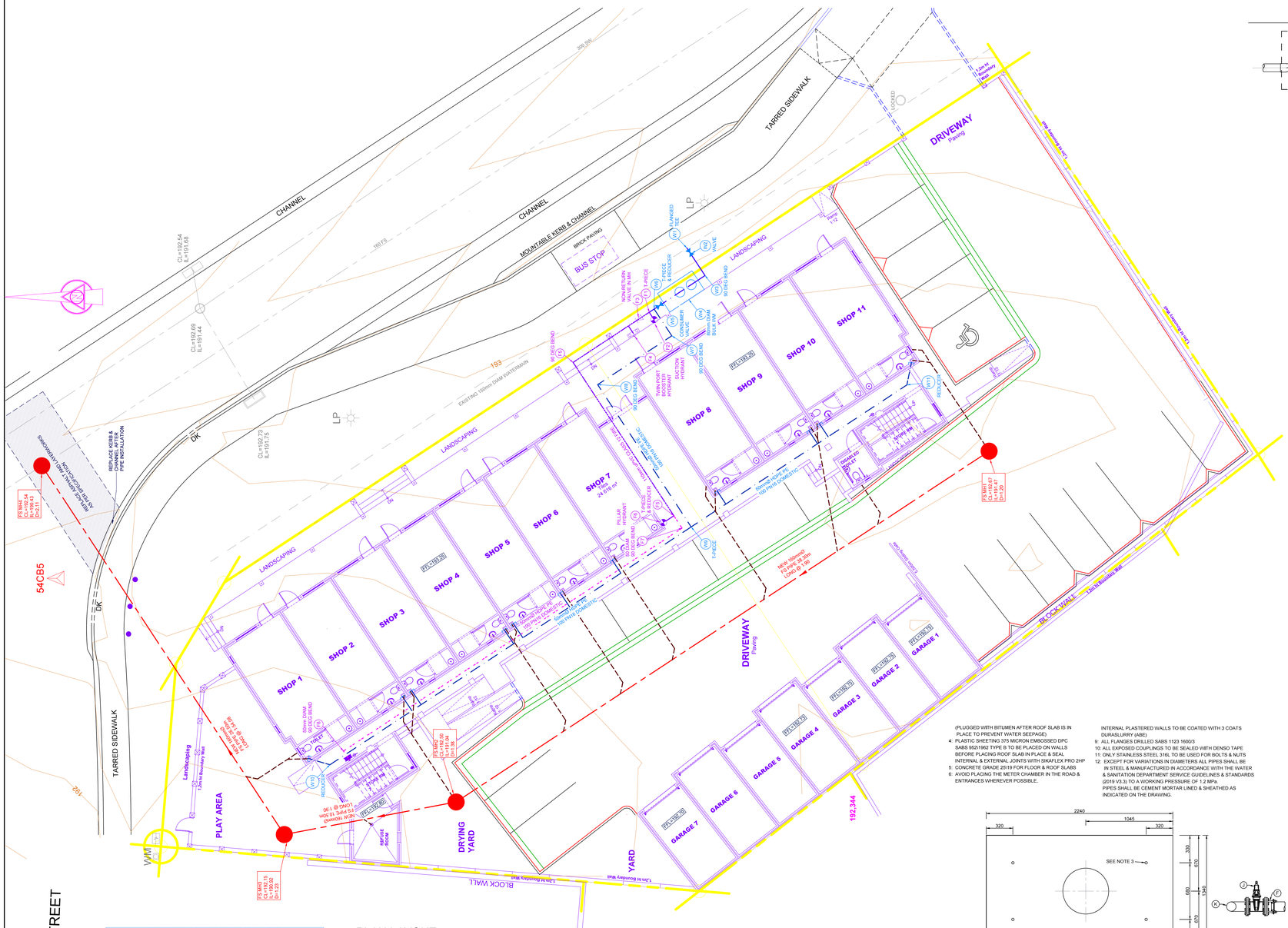
DRIVEWAY
Paving
TARRED SIDEWALK
CHANNEL
MOUNTABLE KERB & CHANNEL
BRICK PAVING
BUS STOP
LANDSCAPING
SHOP 1
SHOP 2
SHOP 3
SHOP 4
SHOP 5
SHOP 6
SHOP 7
24.6169 m²
SHOP 8
SHOP 9
SHOP 10
SHOP 11
GARAGE 1
GARAGE 2
GARAGE 3
GARAGE 4
GARAGE 5
GARAGE 6
GARAGE 7
PLAY AREA
DRYING YARD
YARD
BLOCK WALL
EXISTING 150mm DIAM WATERMAIN
NEW 160mm Ø PVC CLASS 34 FOUL SEWER PIPE
NEW 110mm Ø PVC CLASS 34 FOUL SEWER PIPE
NEW 50mm Ø HDPE PE100 PN16 WATER PIPE
NEW 20mm Ø HDPE PE100 PN16 WATER PIPE
NEW 50mm Ø HDPE PE100 PN16 HOSE REEL MAIN
NEW 110mm Ø PVC CLASS 16 FIRE PIPE
NEW FOUL SEWER MANHOLE
NEW 160mm Ø PVC CLASS 34 FOUL SEWER PIPE
NEW 110mm Ø PVC CLASS 34 FOUL SEWER PIPE
NEW 50mm Ø HDPE PE100 PN16 WATER PIPE
NEW 20mm Ø HDPE PE100 PN16 WATER PIPE
NEW 50mm Ø HDPE PE100 PN16 HOSE REEL MAIN
NEW 110mm Ø PVC CLASS 16 FIRE PIPE
EXISTING SEWER PIPE
EXISTING SEWER MANHOLE
EXISTING STORMWATER PIPE
EXISTING STORMWATER MANHOLE
EXISTING STORMWATER CATCHPIT
EXISTING WATER PIPE
EXISTING FIRE HYDRANT
EXISTING VALVE
HYDRANT OR TWIN BOOSTER SEE FIRE PLAN FOR SPECIFICATIONS
20 MPa CONCRETE PEDESTAL TO BE PAINTED GOLDEN YELLOW BS 356
GMS PIPE CAST IN, Ø80mm FOR HYDRANT, Ø100 FOR BOOSTER - CONFIRM ON FIRE PLAN
700x700x700 20 MPa CONCRETE THRUSTBLOCK IF PIPEWORK NOT THREADED/FLANGED ALL THE WAY TO THE TEE ON THE SUPPLY MAIN
STANDARD 90° BEND, GMS FEMALE THREADED & DENSO TAPED WHERE EXPOSED
OCTAGONAL TOP SIDES = 83 mm
20 MPa CONCRETE PEDESTAL TO BE PAINTED GOLDEN YELLOW BS 356
NGL
NGL
305
355
200
900
350
54CB5
CL=192.54
IL=191.08
CL=192.69
IL=191.44
CL=192.73
IL=191.75
FS 54/4
CL=192.43
IL=190.43
D=2.11
FS 54/3
CL=192.20
IL=190.20
D=1.90
FS 54/2
CL=192.20
IL=189.94
D=1.74
FS 54/1
CL=192.27
IL=189.77
D=1.50
FS 54/5
CL=192.15
IL=190.15
D=1.23
REFLACE KERB & CHANNEL AFTER PIPE INSTALLATION
NOTCH KERB SLIGHTLY TO ENSURE SMOOTH FLOW
SWARMAN AVENUE
3STREET
55
4
DK
LP
FACADE
FENCE
GATE
LOCKED
1.2m Boundary Wall
Lapa
Landscaping
1.2m Boundary Wall
Retaining Wall
Retaining Wall
Retaining

SANS 10400-XA (2021) Instructions for use of this Template	This template is a guide to the Architectural Professional, to assist in complying with the provisions of SANS 10400-XA (2021). © Peter Wyness 2022. The user of this template remains solely responsible for the correct input of data, and the correct functioning of the Excel calculations

SIZE AND LOCATION OF THE BUILDING								
SHOP 1		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					30,36	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		30,36	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENESTRN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D07	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	5,20	0,66	0,49	0,66	1,40	0,29	0,62
	W20	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	5,20	0,66	0,49	0,66	2,06	0,36	1,11
	W02	W,NW,N,NE,E	490	1490	2	1,46	2100	0,54	1134	100	0	5,20	0,66	0,49	0,49	0,72	0,33	0,48
	W03	W,NW,N,NE,E	490	990	1	0,49	2100	0,54	1134	100	0	5,20	0,66	0,49	0,49	0,24	0,33	0,16
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					7,19	→						↓	Limit solar heat gain ∑shgc		4,41	Design ∑shg	2,37
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	5,20	Divide by	W,NW,N,NE,E area sub-total		7,19	and again	7,19
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	5,20	Limit weighted by area average shgc			0,61	Act'l (design) average shgc	0,329
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	5,20						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	5,20						
Storey total fenestration area						7,19												
Storey net internal floor area b /f						30,36												
Fenestration %: IF >20%, use this to enter Table 4						23,67	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 2		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 3		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 4		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENESTRN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 5		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 6		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 7		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 8		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 9		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓

5.1 BUILDING & FENESTRN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 10		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					28,09	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		28,09	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D04	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
	W01	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W05	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,47												
Storey net internal floor area b /f						28,09												
Fenestration %: IF >20%, use this to enter Table 4						19,48	%											

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SIZE AND LOCATION OF THE BUILDING								
SHOP 11		City	ERF 611 MAMRE	Storey	GROUND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					30,36	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		30,36	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	D07	W,NW,N,NE,E	890	2390	1	2,13	2400	0,54	1296	2000	1	5,20	0,66	0,49	0,66	1,40	0,29	0,62
	W20	W,NW,N,NE,E	1490	2090	1	3,11	2400	0,54	1296	2000	1	5,20	0,66	0,49	0,66	2,06	0,36	1,11
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	5,20	0,66	0,49	0,66	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					5,24	→						↓	Limit solar heat gain ∑shgc		3,46	Design ∑shg	1,73
	W08	SW,S,SE	890	1490	1	1,33	n/a	n/a	n/a	n/a	n/a	5,20	Divide by	W,NW,N,NE,E area sub-total		5,24	and again	5,24
	W21	SW,S,SE	490	1490	1	0,73	n/a	n/a	n/a	n/a	n/a	5,20	Limit weighted by area average shgc			0,66	Act'l (design) average shgc	0,329
	W22	SW,S,SE	490	990	1	0,49	n/a	n/a	n/a	n/a	n/a	5,20						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	5,20						
				Storey total fenestration area		7,78												
				Storey net internal floor area b /f		30,36												
Fenestration %: IF >20%, use this to enter Table 4						25,64	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 1		City	ERF 611 MAMRE	Storey	FIRST	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					54,71	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		54,71	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W18	W,NW,N,NE,E	490	1490	1	0,73	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W10	W,NW,N,NE,E	490	990	1	0,49	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W11	W,NW,N,NE,E	490	990	1	0,49	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,81	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	D05	SW,S,SE	890	2115	1	1,88	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,81	and again	6,81
	W15	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						10,11												
Storey net internal floor area b /f						54,71												
Fenestration %: IF >20%, use this to enter Table 4						18,48	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 2		City	ERF 611 MAMRE	Storey	FIRST	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					58,44	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		58,44	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	3660	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						8,64												
Storey net internal floor area b /f						58,44												
Fenestration %: IF >20%, use this to enter Table 4						14,79	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 3		City	ERF 611 MAMRE	Storey	FIRST	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					56,30	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		56,30	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	W17	W,NW,N,NE,E	890	990	4	3,52	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					3,52	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	2	0,46	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		3,52	and again	3,52
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						5,40												
Storey net internal floor area b /f						56,30												
Fenestration %: IF >20%, use this to enter Table 4						9,59	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 4		City	ERF 611 MAMRE	Storey	FIRST	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					58,44	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		58,44	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	3660	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
						Storey total fenestration area												
						Storey net internal floor area b /f												
						Fenestration %: IF >20%, use this to enter Table 4												

Storey total fenestration area	8,64	
Storey net internal floor area b /f	58,44	
Fenestration %: IF >20%, use this to enter Table 4	14,79	%

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SIZE AND LOCATION OF THE BUILDING								
UNIT 5		City	ERF 611 MAMRE	Storey	FIRST	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					54,71	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		54,71	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W09	SW,S,SE	890	1490	1	1,33	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W10	SW,S,SE	490	990	1	0,49	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
	W11	SW,S,SE	490	990	1	0,49	n/a	n/a	n/a	n/a	n/a	0,00						
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00						
						Storey total fenestration area												
						Storey net internal floor area b /f												
Fenestration %: IF >20%, use this to enter Table 4						19,57	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 6		City	ERF 611 MAMRE	Storey	SECOND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					54,71	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		54,71	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W18	W,NW,N,NE,E	490	1490	1	0,73	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W10	W,NW,N,NE,E	490	990	1	0,49	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	W11	W,NW,N,NE,E	490	990	1	0,49	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,81	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	D05	SW,S,SE	890	2115	1	1,88	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,81	and again	6,81
	W15	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						10,11												
Storey net internal floor area b /f						54,71												
Fenestration %: IF >20%, use this to enter Table 4						18,48	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 7		City	ERF 611 MAMRE	Storey	SECOND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					58,44	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		58,44	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	3660	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						8,64												
Storey net internal floor area b /f						58,44												
Fenestration %: IF >20%, use this to enter Table 4						14,79	%											

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SIZE AND LOCATION OF THE BUILDING								
UNIT 8		City	ERF 611 MAMRE	Storey	SECOND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					56,30	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		56,30	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	W17	W,NW,N,NE,E	890	990	4	3,52	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					3,52	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	2	0,46	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		3,52	and again	3,52
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
						Storey total fenestration area												
						Storey net internal floor area b /f												
						Fenestration %: IF >20%, use this to enter Table 4												

Storey total fenestration area	5,40
Storey net internal floor area b /f	56,30
Fenestration %: IF >20%, use this to enter Table 4	9,59 %

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Instructions for use of this Template	The user of this template remains solely responsible for the correct input of data, and the correct functioning of the Excel calculations

SIZE AND LOCATION OF THE BUILDING								
UNIT 9		City	ERF 611 MAMRE	Storey	SECOND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑ →
2- Enter maximum internal building dimension, and multiply out.
Enter dimensions of recesses, rebates, or other negative areas, and deduct....
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					58,44	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		58,44	

LIMITS CALCULATION ↓ COMPLIANCE

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	3660	1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W16	SW,S,SE	390	590	1	0,23	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
		SW,S,SE				0,00	n/a	n/a	n/a	n/a	n/a	0,00						
Storey total fenestration area						8,64												
Storey net internal floor area b /f						58,44												
Fenestration %: IF >20%, use this to enter Table 4						14,79	%											

SANS 10400-XA (2021)
Instructions for use of this Template

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SIZE AND LOCATION OF THE BUILDING								
UNIT 10		City	ERF 611 MAMRE	Storey	SECOND	Latitude		33,9

1- Enter project name ↑ city ⌂ storey ↑ & latitude ↑
2- Enter maximum internal building dimension, and multiply out. →
Enter dimensions of recesses, rebates, or other negative areas, and deduct.... →
3- Transfer total to second lowest cell of Coumn H

max intnl dimension		e-w	n-s	number	over'll area	Other occupancies
Positive spaces					54,71	
					0,00	
minus negative spaces					0,00	
	rebate				0,00	
	recess				0,00	
	other				0,00	
Use a separate page for larger plans			Total storey net internal floor area c/f		54,71	

LIMITS CALCULATION ↓

5.1 BUILDING & FENEST'RN ORIENTATION			FENESTRATION AREAS				SHADING - Clause 5.2					U & shgc VALUES - CLAUSE 5.3				Averaging	COMPLIANCE	
Catalogue reference	window plan ref	Orientation sector	window width	window vertical	number (quantity)	window area m²	shading height	multi-plier (Table 3)	required shading projection	actual shdg (design) projection	compliant 1= yes 0= no	U-value	Enter shgc compliant T4, Col 3	Enter non-compliant shgc T4 Col 4	required shgc-value From cols O&P	ww area x required shgc	Actual (design)shgc specified	area x actual (design) shgc
	SD01	W,NW,N,NE,E	1790	2115	1	3,79	2125	0,54	1147,5	2660	1	0,00	0,00	0,00	0,00	0,00		0,00
	W13	W,NW,N,NE,E	890	1490	1	1,33	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
	D05	W,NW,N,NE,E	890	2115	1	1,88	2125	0,54	1147,5	100	0	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
		W,NW,N,NE,E				0,00		0,54	0		1	0,00	0,00	0,00	0,00	0,00		0,00
	Total W,NW,N,NE,E orientated fenest'n area					6,99	→						↓	Limit solar heat gain ∑shgc		0,00	Design ∑shg	0,00
	W09	SW,S,SE	890	1490	1	1,33	n/a	n/a	n/a	n/a	n/a	0,00	Divide by	W,NW,N,NE,E area sub-total		6,99	and again	6,99
	W10	SW,S,SE	490	990	1	0,49	n/a	n/a	n/a	n/a	n/a	0,00	Limit weighted by area average shgc			0,00	Act'l (design) average shgc	0,000
	W11	SW,S,SE	490	990	1	0,49	n/a	n/a	n/a	n/a	n/a	0,00						
	W14	SW,S,SE	1190	1190	1	1,42	n/a	n/a	n/a	n/a	n/a	0,00						
				Storey total fenestration area		10,71												
				Storey net internal floor area b /f		54,71												
Fenestration %: IF >20%, use this to enter Table 4						19,57	%											